

MdKIN10-mediated phosphorylation of the E3 ubiquitin ligase MdMIEL1 leads to its autophagic degradation under cold stress

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Short title: Phosphorylation is key in MdMIEL1 degradation

Abstract

MYB30-Interacting E3 Ligase1 (MdMIEL1) plays a negative role in apple (*Malus × domestica*) cold tolerance by mediating the degradation of its target proteins. Its protein stability decreases under cold stress; however, how MdMIEL1 is regulated under cold conditions remains unclear. Here, we report that the α -catalytic subunit of sucrose non-fermenting-1-related protein kinase, MdKIN10, a positive regulator of cold stress, interacts with MdMIEL1, leading to its phosphorylation and subsequent degradation. Specifically, cold-activated MdKIN10 phosphorylates MdMIEL1 at serine 198, which occurs near the AIM motif; this modification further enhances the interaction between MdMIEL1 and MdATG8i, thereby promoting MdMIEL1 degradation through the autophagy pathway. Furthermore, MdBBX7, which functions as a positive regulator in

apple cold stress responses by positively regulating the expression of *ω-3 fatty acid desaturase (MdFAD8)* and *C-repeat-binding factor 2 (MdCBF2)*, thus promoting fatty acid desaturation and cold-responsive gene expression, is a target of MdMIEL1. In addition, MdKIN10-mediated phosphorylation of MdMIEL1 at Ser198 attenuates the ability of MdMIEL1 to inhibit apple cold tolerance and decrease MdBBX7 protein abundance in response to cold stress. Our study reveals an intricate post-translational modification cascade of MdKIN10-MdMIEL1-MdBBX7 molecular module under cold stress in apple.

Introduction

Protein degradation serves a dual role in biological systems, not only enabling the clearance of malfunctioning or surplus proteins but also playing a crucial regulatory role in various developmental and stress response processes (Dikic 2017). The ubiquitin-proteasome system (UPS) and autophagy are the two principal degradation pathways in eukaryotes. The UPS comprises ubiquitin (Ub), ubiquitin-activating enzymes (E1s), ubiquitin-conjugating enzymes (E2s), ubiquitin ligases (E3s), proteasomes and their substrates (Lee and Kim 2011). Within this system, E3 plays a pivotal role in determining the specificity of protein substrates for degradation (Stone and Callis 2007). Autophagy is a highly conserved intracellular degradation system in eukaryotes that utilizes double-layered vesicles called autophagosomes to mediate the degradation of unnecessary or dysfunctional cytoplasmic components in vacuoles (in yeasts and plants) or lysosomes (in animals) (Han et al. 2011). This process is coordinated by a series of autophagy-related proteins (ATGs) that function at different stages of autophagy. Over forty ATG proteins have been characterized across yeast, animals, and plants (Marshall and Vierstra 2018). Notably, ATG8 is an essential component in the selective cargo recognition and autophagosome assembly process (Noda et al. 2010). Additionally, ATG5 is crucial for the assembly of autophagosomes by anchoring ATG8 to the phagosome membrane (Thompson et al. 2005).

The RING-type E3 ligase MYB30-Interacting E3 Ligase1 (MIEL1) is a ubiquitin E3 ligase that often degrades its substrate *via* the 26-dependent proteasome pathway. So far, the role of MIEL1 in plant development, abiotic stress, and biotic stress responses

has been extensively investigated in both *Arabidopsis* (*Arabidopsis thaliana*) and apple (*Malus × domestica*) (Marino et al. 2013; Lee and Seo 2016; Zhao and Xue 2020). In *Arabidopsis*, MIEL1 degrades MYB96 and MYB30, thereby playing a negative role in wax biosynthesis, seed germination, and root development (Lee and Seo 2016; Gil et al. 2017; Zhao and Xue 2020). In apple, MdMIEL1 acts as a negative regulator of cold and drought stress tolerance by promoting the degradation of MdMYB1, MdMYB308L, and MdBBX37 under cold stress, and MdBBX7 under drought stress (An et al. 2017, 2020, 2021; Chen et al. 2021). Collectively, MIEL1 modulates protein stability by mediating ubiquitination and proteasomal degradation of its target proteins. Therefore, the protein stability of MIEL1 must be precisely controlled to ensure normal plant development and appropriate responses to various stresses. Interestingly, studies reported that the stability of the MIEL1 protein in apple is reduced in response to cold and drought (An et al. 2020; Chen et al. 2022). However, the mechanisms by which MIEL1 is degraded in various biological processes remain largely unknown.

Alongside ubiquitination, phosphorylation stands out as another widely investigated post-translational modification (PTM) that governs protein stability. Whether ubiquitination or phosphorylation, the protein degradation or stabilization they govern is primarily mediated by the UPS. SnRK1 is a heterotrimeric complex consisting of an α -catalytic (KIN10, KIN11, and KIN12 in *Arabidopsis*) and two regulatory (β and plant-specific $\beta\gamma$) subunits (Emanuelle et al. 2015). The catalytic α subunit contributes to the kinase activity of this complex, with their catalytic activity absolutely depends on the phosphorylation of a specific threonine residue in the kinase domain T-loop by its upstream kinases SnRK1 Activating Kinase 1 and 2 (SnAK1/GRIK1 and SnAK2/GRIK2) (Shen et al. 2009). SnRK1 serves as a central metabolic regulator of cellular energy homeostasis and has been shown to play a key role in plant growth and development as well as biotic and abiotic stress tolerance (Chan et al. 2017; Han et al. 2020; Sun et al. 2021). For instance, KIN10 phosphorylates NLP7, targeting it for proteasomal degradation, which in turn inhibits nitrate-mediated gene expression and plant growth under conditions of carbon deficiency or nitrate depletion (Wang et al. 2022). Similarly, OsSnRK1a mediates phosphorylation and degradation of OsNAC23 by the 26S

1 proteasome-mediated pathway in rice (*Oryza sativa*) (Li et al. 2022). In barley (*Hordeum*
2 *vulgare*), SnRK1 enhances resistance to powdery mildew by phosphorylating and
3 destabilizing WRKY3 through the proteasome pathway (Han et al. 2020). However,
4 whether SnRK1 mediates phosphorylation and degradation of substrate *via* autophagy
5 pathway is unclear.

6 Cold stress, including chilling (0-15°C) and freezing (< 0°C), affects the
7 geographical distribution, growth, development, and yield of plants. One of the main
8 consequences of cold temperature injury is a decrease in the integrity and fluidity of the
9 cell membrane, thereby compromising the function of the cell membrane (Murata and
10 Los 1997). Generally, to maintain the correct membrane fluidity, the degree of
11 unsaturation of fatty acids in their membrane increases, a process catalyzed by a group
12 of fatty acid desaturases (FADs) (Sato and Murata 1980). Extensive research has
13 demonstrated that the upregulation of *FADs*, such as *FAD2*, *FAD3*, *FAD6*, *FAD7*, and
14 *FAD8*, enhances cold tolerance in plants (Gombos et al. 1992; Domínguez et al. 2010;
15 Tovuu et al. 2016; Wang et al. 2021a). The apple trees are vulnerable to freezing
16 temperatures, especially in the flowering stage, leading to decreased quality and yield
17 (Farajzadeh et al. 2010). Many plants including the apple trees have evolved complex
18 strategies to withstand cold stress, one of which is cold acclimation, a process by which
19 plants acquire increased freezing tolerance through prior exposure to low but non-
20 freezing temperatures (Guy 1990; Thomashow 1999). Under cold stress, expression of
21 *C-REPEAT BINDING FACTOR/DE-HYDRATION RESPONSIVE ELEMENT-BINDING*
22 *PROTEIN1* (*CBF/DREB1*) is rapidly induced, thereby activating the expression of
23 *COLD-RESPONSIVE* (*COR*) genes and improving cold tolerance (Yamaguchi-
24 Shinozaki and Shinozaki 1994; Stockinger et al. 1997).

25 Here, we demonstrated that MdKIN10-mediated phosphorylation of MdMIEL1 is
26 crucial for the interaction between MdMIEL1 and MdATG8i, as well as for the cold
27 stress-induced degradation of MdMIEL1 in apple. The degradation of MdMIEL1 enables
28 the release of MdBBX7, the activator of *MdFAD8* and *MdCBF2*, leading to enhanced
29 PUFA percentage and cold-responsive gene expression under cold stress, thereby
30 enhancing freezing tolerance in apple. Thus, our results not only discover a complex

1 'cascade' of various PTMs of MdKIN10-MdMIEL1-MdBBX7 molecular module under
2 cold stress, but also demonstrate that phosphorylation plays an essential role in the
3 autophagic degradation of proteins in plants.

4 5 **Results**

6 **MdKIN10 interacts with MdMIEL1 *in vivo* and *in vitro***

7 MdMIEL1 is a critical negative regulator of cold and drought stress signaling; therefore,
8 its protein stability must be precisely controlled to ensure normal plant development and
9 appropriate responses to various stresses (An et al. 2020; Chen et al. 2022). To explore
10 the regulation factors of MdMIEL1 itself, we conducted a yeast two-hybrid (Y2H) screen
11 to identify MdMIEL1-interacting proteins using MdMIEL1 as bait. Among the identified
12 interactors, a homolog protein of Arabidopsis SNF1 KINASE HOMOLOG 10 (SnRK1.1)-
13 MdKIN10 was identified. We confirmed the interaction between MdKIN10 and MdMIEL1
14 in yeast cells using the Y2H assay (Fig. 1A). To determine whether this interaction
15 occurs in plants, we performed a split luciferase complementation assay (Split-LUC).
16 The results showed a strong luminescence signal during the co-expression of
17 MdMIEL1-nLUC and cLUC-MdKIN10 (Fig. 1B). Furthermore, co-immunoprecipitation
18 (Co-IP) experiments revealed that MdKIN10 tagged with a green fluorescent protein
19 (GFP) interacted with MdMIEL1 fused to a FLAG tag in *Nicotiana benthamiana* leaves,
20 while no interaction was detected between GFP and MdMIEL1 (Fig. 1C). Finally, strong
21 fluorescent signals were observed in the endoplasmic reticulum (ER) of *N. benthamiana*
22 leaf epidermal cells when co-transformed with MdKIN10-nYFP and MdMIEL1-cYFP
23 (Fig. 1D). Collectively, these results indicate that MdKIN10 interacts with MdMIEL1 both
24 *in vitro* and *in vivo*.

25 **MdKIN10 phosphorylates and destabilizes MdMIEL1 in response to cold**

26 Given that MdMIEL1 functions as an E3 ligase responsible for the degradation of target
27 proteins, we investigated whether MdKIN10 was one of its targets. However, the results
28 demonstrate that MdMIEL1 did not influence the protein stability of MdKIN10
29 (Supplementary Fig. S1A). Surprisingly, when we co-expressed MdMIEL1-FLAG and

1 MdKIN10-GFP or GFP in *N. benthamiana* leaves, we found that MdKIN10 could
2 facilitate the degradation of MdMIEL1 (Supplementary Fig. S1B).

3 Considering that MdKIN10 is the catalytic subunit of the SnRK1 kinase complex
4 and facilitates the degradation of MdMIEL1, we hypothesized that MdMIEL1 might be a
5 substrate of MdKIN10. We then conducted an *in vitro* kinase assay to determine
6 whether MdKIN10 phosphorylated MdMIEL1. MdKIN10 containing the S_TKc and UBA
7 domains fused with Maltose binding protein (MBP) tag was purified due to the
8 insolubility of the full-length KIN10 protein (Supplementary Fig. S1C). Previous research
9 has shown that the kinase activity of KIN10 relies on its phosphorylation by the
10 upstream kinases GRIK1/GRIK2 (Wang et al. 2021b, 2022). The result showed that
11 MdKIN10 was phosphorylated when co-incubated with MdGRIK2 (Supplementary Fig.
12 S1D). Furthermore, MdKIN10-KD-MBP phosphorylated MdMIEL1-MBP in the presence
13 of MdGRIK2-HIS, while MdGRIK2-HIS alone did not phosphorylate MdMIEL1-MBP,
14 indicating that MdMIEL1 phosphorylation is dependent on MdKIN10 (Fig. 2A). To
15 identify the phosphorylation sites of MdMIEL1, we conducted LC-MS/MS analysis. It
16 was shown that Ser198 in MdMIEL1 protein was potentially phosphorylated (Fig. 2B).
17 To demonstrate whether this residue is the target site for MdMIEL1 phosphorylation by
18 MdKIN10, we generated a mutant MdMIEL1 protein where Ser198 was substituted with
19 alanine (MdMIEL1^{S198A}) to prevent phosphorylation. Subsequent kinase assay
20 demonstrated that this substitution diminishes MdKIN10-mediated phosphorylation of
21 the MdMIEL1 (Fig. 2C).

22 MdMIEL1 has been shown to be largely degraded under cold and drought
23 conditions (An et al. 2020; Chen et al. 2022). To ascertain whether MdKIN10 is involved
24 in this observation, we first examined the phosphorylation status of MdKIN10 under cold
25 and drought conditions. Immunoblot analysis revealed a marked and rapid increase in
26 MdKIN10 activity following cold treatment (Fig. 2D). In contrast, a significant decrease in
27 MdKIN10 activity was observed under drought treatment (Supplementary Fig. S1E). We
28 proceeded to assess the impact of MdKIN10 on MdMIEL1 stability *in vivo* under cold or
29 drought stress. To this end, we generated transgenic calli overexpressing
30 35S:MdMIEL1-FLAG (MdMIEL1 OE) or co-overexpressing 35S:MdKIN10 and

35S:MdMIEL1-FLAG (*MdKIN10* OE/*MdMIEL1* OE). Results showed that MdMIEL1 protein abundance was decreased in response to cold treatment in the apple (Fig. 2E). Importantly, the MdMIEL1 degradation was significantly enhanced by the overexpression of *MdKIN10* following cold treatment (Fig. 2E). However, comparable MdMIEL1 degradation was observed in both *MdMIEL1* OE and *MdKIN10* OE/*MdMIEL1* OE transgenic calli under drought treatment, as shown in Supplementary Fig. S1F. These results suggest that MdKIN10 might be responsible for the degradation of MdMIEL1 under cold, but not under drought conditions. Therefore, we focus on how MdKIN10 regulates MdMIEL1 protein stability under cold conditions.

To assess MdMIEL1 degradation is dependent on MdKIN10-mediated phosphorylation, we generated *MdMIEL1*^{S198A} OE and *MdMIEL1* OE transgenic calli with similar levels of *MdMIEL1* expression (Supplementary Fig. S1G). The results reveal that cold treatment significantly accelerated the degradation of MdMIEL1 in *MdMIEL1* OE but had minimal effects on the MdMIEL1^{S198A} in *MdMIEL1*^{S198A} OE (Fig. 2F), suggesting that MdKIN10 phosphorylates MdMIEL1 at Ser198 and facilitates its degradation. Given that protein phosphorylation can influence subcellular localization (Prak et al. 2008; Wang et al. 2022, 2023), we investigated whether MdKIN10-mediated phosphorylation affects the subcellular localization of MdMIEL1. Results showed that fluorescent signals of MdMIEL1, MdMIEL1^{S198A}, and MdMIEL1^{S198D} were localized in both the nucleus and cytoplasm (Supplementary Fig. S1H). These findings suggest that MdKIN10 does not alter the subcellular localization of MdMIEL1.

To further verify that Ser198 is the phosphorylation site of MdMIEL1 *in planta* in response to cold, we analyzed the phosphorylation levels of MdMIEL1 in *MdMIEL1* OE, *MdMIEL1*^{S198A} OE and *MdKIN10* OE/*MdMIEL1* OE transgenic calli during a time-course cold treatment. MdMIEL1 was immunoprecipitated with an anti-FLAG antibody, and the phosphorylation level of MdMIEL1 was detected using a P-Ser antibody. We observed that the phosphorylation level of MdMIEL1 was significantly increased in *MdKIN10* OE/*MdMIEL1* OE, while notably decreased in *MdMIEL1*^{S198A} OE and *MdMIEL1* OE transgenic calli before cold treatment (Fig. 2G). Moreover, the phosphorylation of MdMIEL1 was markedly elevated in *MdKIN10* OE/*MdMIEL1* OE and *MdMIEL1* OE calli

1 after exposure to cold conditions, as compared to the *MdMIEL1* OE transgenic calli (Fig.
2 2G). However, the MdMIEL1 phosphorylation level in *MdMIEL1*^{S198A} OE transgenic calli
3 was barely induced by cold treatment compared to that in *MdMIEL1* OE transgenic calli
4 (Fig. 2G). Taken together, these results demonstrate that the cold-induced
5 phosphorylation of Ser198 on MdMIEL1 is strictly dependent on MdKIN10.

6 To determine whether MdKIN10-mediated Ser198 phosphorylation of MdMIEL1
7 might be conserved across angiosperms, we first investigated the interaction domains
8 of MdKIN10 and MdMIEL1. Y2H assays revealed that the zf-CHY domain of MdMIEL1
9 and the KA1 domain of MdKIN10 are responsible for their interaction in yeast
10 (Supplementary Fig. S2A). Furthermore, we examined homologous MdMIEL1 and
11 MdKIN10 proteins from various eudicots and monocots, and found that these critical
12 domains are highly conserved (Supplementary Fig. S2B). Subsequently, we analyzed
13 the conservation of the Ser198 residue across these species. Our analysis revealed that
14 Ser198 is fully conserved in eudicots and partially conserved in monocots
15 (Supplementary Fig. S2C). Notably, Ser198 is replaced by threonine in some monocots,
16 which nonetheless serves as the target phosphorylation residue of MdKIN10 (Weekes
17 et al. 1993). Together, these results indicate that MdKIN10-mediated Ser198
18 phosphorylation of MdMIEL1 is mostly conserved in angiosperm.

19 **Phosphorylation of MdMIEL1 is involved in the regulation of apple cold tolerance**

20 We next investigated whether MdMIEL1 plays a regulatory role in apple cold tolerance.
21 The results reveal that, regardless of cold acclimation, transgenic apple plants with
22 reduced *MdMIEL1* expression (*MdMIEL1* RNAi) displayed enhanced survival rates and
23 reduced leaf electrolyte leakage compared to non-transgenic plants (GL-3, the genetic
24 background for all transgenic plants) (Supplementary Fig. S3A-C). In contrast, plants
25 overexpressing *MdMIEL1* (*MdMIEL1* OE) exhibited hypersensitivity to freezing stress,
26 as indicated by lower survival rates and higher leaf electrolyte leakage (Supplementary
27 Fig. S3D-F). These results demonstrate that MdMIEL1 functions as a negative regulator
28 of freezing tolerance in apple.

29 To elucidate the impact of MdKIN10-mediated phosphorylation on MdMIEL1
30 function under cold stress, besides *MdMIEL1* OE and *MdMIEL1*^{S198A} OE calli, we also

1 mutated the Ser198 of MdMIEL1 to aspartic acid (D) to mimic a phosphorylated state
2 and generated transgenic calli expressing *35S:MdMIEL1^{S198D}-FLAG* (*MdMIEL1^{S198D}*
3 OE). Transgenic calli with similar expression levels of *MdMIEL1* in *MdMIEL1* OE,
4 *MdMIEL1^{S198D}* OE, and *MdMIEL1^{S198A}* OE were selected for further analysis
5 (Supplementary Fig. S1H). As shown in Fig. 2H and 2I, following 10 days of cold
6 treatment, all transgenic calli showed slower growth rates compared to the wild type
7 (WT). However, compared with the *MdMIEL1* OE calli, this reduction is significantly
8 alleviated in *MdMIEL1^{S198D}* OE calli, while it is notably exacerbated in *MdMIEL1^{S198A}* OE
9 calli. These results suggest that MdKIN10-mediated phosphorylation of MdMIEL1 at
10 Ser198 attenuates the inhibitory effect of MdMIEL1 on apple cold tolerance.

11 **MdKIN10 is required for the cold stress response in apple**

12 Cold treatment has been shown to induce MdKIN10 kinase activity, which in turn
13 promotes the degradation of MdMIEL1, suggesting a role for MdKIN10 in cold stress
14 response. We found that *MdKIN10* were induced within the first three hours of cold
15 treatment and were then slightly repressed by prolonged cold stress in leaves
16 (Supplementary Fig. S4A), suggesting that transcriptional regulation is not the primary
17 factor contributing to the cold-induced kinase activity of MdKIN10 (Fig. 2D). Subcellular
18 localization indicates that MdKIN10 is localized to the nucleus and ER (Supplementary
19 Fig. S4B). Phylogenetic analysis showed that KIN10 orthologs are conserved across all
20 plant species, from chlorophyta to eudicots (Supplementary Fig. S4C).

21 To elucidate the biological role of MdKIN10 in cold stress, we generated *MdKIN10*
22 OE and RNAi transgenic plants in GL-3 (Supplementary Fig. S4D) (Dai et al. 2013).
23 After freezing stress, the *MdKIN10* RNAi plants exhibited a lower survival rate
24 compared to the GL-3 (Fig. 3A, B), while the overexpressing of *MdKIN10* enhanced the
25 survival rate of apple (Fig. 3D, E). Electrolyte leakage assays further demonstrated that
26 the *MdKIN10* RNAi plants were more sensitive to freezing stress (Fig. 3C), whereas
27 *MdKIN10* OE plants improved tolerance to freezing stress (Fig. 3F). These results
28 collectively indicate that MdKIN10 is a key factor in improving the freezing tolerance of
29 apple plants.

To further explore the genetic relationship between MdKIN10 and MdMIEL1 under cold stress, we generated *MdKIN10* OE transgenic calli and subjected them to cold treatment along with WT, *MdMIEL1* OE, and *MdKIN10* OE/*MdMIEL1* OE transgenic calli. After cold stress, the *MdKIN10* OE calli showed a higher relative growth rate compared to the wild type, while the *MdMIEL1* OE calli exhibited a lower relative growth rate. Notably, the relative growth rate of the *MdKIN10* /*MdMIEL1* OE calli was comparable to that of the *MdMIEL1* OE calli (Supplementary Fig. S5A, B). These results suggest that MdMIEL1 functions downstream of MdKIN10 in the cold stress response of apple.

MdKIN10 promotes autophagic degradation of MdMIEL1 in response to cold stress

To elucidate whether MdKIN10 mediates the degradation of MdMIEL1 via the proteasomal or autophagy pathway, we co-expressed MdMIEL1-FLAG and MdKIN10-GFP or GFP alone in *N. benthamiana* leaves with the presence or absence of the proteasome inhibitor MG132 or the autophagy inhibitor E64d. Overexpression of *MdKIN10* triggered the degradation of the MdMIEL1 protein, and this degradation was markedly attenuated by E64d but not MG132 (Fig. 4A), suggesting that MdKIN10 induced degradation of MdMIEL1 via autophagy rather than the proteasome pathway.

Then we explored whether MdMIEL1 was degraded by MdKIN10 through the autophagy pathway under cold stress. Firstly, we explored the effects of the autophagy inhibitor concanamycin A (ConA) on MdMIEL1 degradation under cold conditions using calli expressing *GFP-MdMIEL1*. In the absence of ConA, we observed a release of free GFP from GFP-MdMIEL1 upon exposure to cold stress. However, in the presence of ConA, no detectable release of free GFP was observed under cold treatment (Fig. 4B). Moreover, upon cold treatment, YFP-MdMIEL1 was observed in small puncta within the vacuolar, and the treatment with ConA, led to an increased accumulation of YFP-MdMIEL1 in these puncta (Fig. 4C, D). Notably, the fluorescence signals of YFP-MdMIEL1 appeared to decrease in both the nucleus and cytoplasm in response to cold. To further validate this observation, we examined the nuclear-cytoplasmic distribution of MdMIEL1 in *35S:MdMIEL1-FLAG* transgenic calli using cell fractionation assays under

control and cold conditions. Immunoblot analysis revealed that the protein levels of MdMIEL1 in both the nucleus and cytoplasm were significantly reduced upon exposure to cold stress (Fig. 4E). We hypothesize that this phenomenon may result from low temperature-induced recruitment of MdMIEL1 into autophagosomes in the cytoplasm for degradation, thereby reducing its nuclear import. To further determine the necessity of the autophagy component in MdMIEL1 degradation in response to cold, we transiently knocked down the autophagy-related gene *MdATG5* and overexpressed *GFP-MdMIEL1* in the calli (Fig. S6A). ATG5 is an essential component of the autophagy machinery (Thompson et al. 2005). Knockdown of *MdATG5* markedly reduced ATG8 lipidation levels in response to cold stress, confirming impaired autophagy (Fig. S6B). Moreover, the release of free GFP from GFP-MdMIEL1 was significantly inhibited when *MdATG5* expression was repressed under control and cold conditions (Fig. 4F). Together, these results suggest that autophagy pathway is necessary for MdMIEL1 degradation under cold conditions. To further elucidate whether MdKIN10 promotes the degradation of MdMIEL1 *via* the autophagy pathway under cold stress, we transiently expressed GFP-MdMIEL1 in the wild type or *MdKIN10* OE calli, and subsequently treated them with ConA and cold stress. The results reveal that in the absence of ConA, a significantly higher accumulation of free GFP was observed in *MdKIN10* OE calli compared to the wild-type calli. However, in the presence of ConA, free GFP was undetectable, and the accumulation of GFP-MdMIEL1 protein levels was comparable between the wild type and *MdKIN10* OE calli (Fig. 4G). These results indicate that MdKIN10 promotes MdMIEL1 degradation *via* the autophagy pathway in response to cold stress.

Ser198 phosphorylation of MdMIEL1 mediated by MdKIN10 is critical for its autophagic degradation

Next, we investigated the regulatory mechanism of MdKIN10-induced MdMIEL1 autophagic degradation. A critical step in autophagosome formation involves the conjugation of the ubiquitin-like protein ATG8 to autophagic membranes, which interacts with specific proteins to facilitate cargo recruitment and phagophore assembly (Johansen and Lamark 2020; Luo et al. 2023). We conducted a Y2H assay to analyze the possible interaction of MdMIEL1 with the three MdATG8s in apple, namely

MdATG8c (the homologs of AtATG8a, AtATG8b, AtATG8c and AtATG8d), MdATG8f (the homologs of AtATG8e, AtATG8f and AtATG8g), and MdATG8i (the homologs of AtATG8h and AtATG8i). The results showed that MdMIEL1 specifically interacts with MdATG8i (Fig. 5A), but not with MdATG8c and MdATG8f (Supplementary Fig. S6C). This interaction was further verified by the Split-LUC assay (Fig. 5B). Generally, proteins that interact with ATG8 contain the ATG8-interacting motif (AIM; W/F/Y-X-X-L/I/V) (Noda et al. 2008, 2010). Using the iLIR software (Kalvari et al. 2014), we found that MdMIEL1 has one high-scoring AIM. We then mutated the key residues of the AIM motif (MdMIEL1^{mAIM}: W206A I209A) to alanine and performed Y2H and pull-down assay between MdMIEL1^{mAIM} and MdATG8i. The results displayed that MdMIEL1^{mAIM} did not interact with MdATG8i (Fig. 5A, C), indicating the essential role of the AIM motif in the MdMIEL1-MdATG8i interaction. To explore the role of the AIM motif in the degradation of MdMIEL1 under cold conditions, we expressed GFP-MdMIEL1 and GFP-MdMIEL1^{mAIM} separately in apple calli and assessed their cleavage in response to cold treatment. The results showed that mutation of the AIM motif in GFP-MdMIEL1 completely abolished free GFP accumulation in calli under both normal and cold conditions. Moreover, the content of GFP-MdMIEL1^{mAIM} protein was significantly higher than that of GFP-MdMIEL1 under all tested conditions. These results indicate that the AIM motif is essential for MdMIEL1 degradation in response to cold stress (Fig. 5D). Notably, the AIM motif is close to the Ser198 phosphorylation site targeted by MdKIN10 (Fig. 5E), we then investigated the influence of Ser198 phosphorylation of MdMIEL1 on the interaction between MdMIEL1 and MdATG8i through a series of assays, including an *in vitro* pull-down assay, as well as two *in vivo* assays: Split-LUC and Co-IP. The results showed that MdMIEL1^{S198D} exhibited an enhanced interaction with MdATG8i (Fig. 5F, H), whereas the MdMIEL1^{S198A} mutant displayed a reduced interaction with MdATG8i compared to the MdMIEL1 (Fig. 5G, H). These results demonstrate that MdKIN10-mediated phosphorylation of Ser198 at MdMIEL1 facilitates the interaction between MdMIEL1 and MdATG8i.

Considering that autophagy commonly utilizes ATG8 to transport target proteins to the autophagosome for degradation (Marshall and Vierstra 2018), we hypothesized that MdMIEL1 might be recruited by MdATG8i to the autophagosome for selective

degradation. To test this hypothesis, we observed the colocalization between MdMIEL1 and MdATG8i without ConA treatment under cold conditions. The results showed strong colocalization of mCherry-MdMIEL1 and YFP-MdATG8i in autophagosomes, indicating that MdMIEL1 is targeted by MdATG8i under cold conditions. In contrast, mCherry-MdMIEL1^{mAIM} failed to co-localize with YFP-MdATG8i, clearly indicating that the AIM motif is essential for MdATG8i-mediated targeting of MdMIEL1 (Fig. 5I; Fig. S6D). We next asked whether Ser198 phosphorylation modulates this targeting. As shown in Fig. 5J and S6E, the mCherry-MdMIEL1^{S198A} exhibited a diminished capacity for colocalization with autophagosomes labeled by YFP-MdATG8i within the cytoplasm, indicating that Ser198 phosphorylation is required for the targeting of MdMIEL1 by MdATG8i. Therefore, Ser198 phosphorylation of MdMIEL1 is crucial for its interaction with MdATG8i and autophagic degradation.

Abiotic stress frequently triggers an increase in autophagic activity (Li et al. 2020b; Che et al. 2023; Huang et al. 2024). We next sought to determine whether cold stress affects autophagy activity. To this end, we firstly assessed autophagic flux by examining the release of free GFP from the GFP-MdATG8i fusion protein in response to cold stress in apple. Upon cold stress treatment, a substantial free GFP was derived from GFP-ATG8i in the calli transiently expressing *GFP-ATG8i* (Supplementary Fig. S7A). The conjugation of ATG8 to phosphatidylethanolamine (ATG8-PE) is another marker of autophagy activation (Chen et al. 2017). Therefore, we examined ATG8 and ATG8-PE abundance in the GL-3 using an anti-ATG8 antibody in response to cold stress. The results showed that the ratio of ATG8-PE to ATG8 increased in response to cold treatment (Supplementary Fig. S7B). Together, the above results demonstrate that cold stress enhances autophagic activity, suggesting that cold-induced autophagic activity may be involved in the autophagic degradation of MdMIEL1 under cold stress.

MdBBX7 functions as a key positive regulator of cold tolerance, operating downstream of MdMIEL1

Previously, we found that MdMIEL1 ubiquitinates and degrades the MdBBX7 to negatively regulate drought stress response in apple (Chen et al. 2022). Hence, we asked whether MdMIEL1 regulates cold tolerance in apple by mediating the

ubiquitination and degradation of MdBBX7. The results showed that MdBBX7 exhibits a higher level of ubiquitination in *MdMIEL1* OE plants compared to GL-3 plants under both normal and cold conditions, with reduced ubiquitination levels of MdBBX7 observed during cold treatment (Supplementary Fig. S8). Furthermore, MdBBX7 was less abundant in *MdMIEL1* OE plants under both control and cold conditions; however, the increased MdBBX7 levels were observed upon treatment with MG132 under cold treatment (Fig. 6A). In conclusion, these results suggest that *MdMIEL1* plays a role in mediating the ubiquitination and degradation of MdBBX7 in response to cold stress.

To explore the function of MdBBX7 in apple cold stress response, we examined the expression of *MdBBX7* under cold conditions, RT-qPCR showed the induced expression of *MdBBX7*, as well as its increased protein accumulation under cold treatment (Supplementary Fig. S9A, B). In addition, we found that *MdBBX7* RNAi plants were hypersensitive to freezing stress (Fig. 6B-D), while *MdBBX7* OE plants showed increased tolerance to freezing stress (Fig. 6E-G). These results suggest that MdBBX7 enhances the cold tolerance of apple.

To further clarify the genetic relationship between *MdMIEL1* and MdBBX7 under cold conditions, we conducted an analysis of the relative growth rate of WT, *MdBBX7* RNAi, *MdMIEL1* RNAi, and *MdMIEL1* anti/*MdBBX7* RNAi calli following cold treatment. *MdMIEL1* RNAi calli exhibited enhanced cold tolerance compared to the WT, while *MdBBX7* RNAi calli displayed reduced cold tolerance. *MdMIEL1* anti/*MdBBX7* RNAi calli showed a significant decrease in cold stress tolerance compared to the *MdMIEL1* RNAi calli (Supplementary Fig. S10A, B). Taken together, these results imply that *MdMIEL1* represses the function of MdBBX7 under cold stress.

MdBBX7 confers cold tolerance through regulating FA desaturation and cold-responsive gene expression

To elucidate the molecular mechanisms underlying MdBBX7-mediated cold tolerance in apple, we performed an RNA-Seq analysis with the leaves of GL-3 and *MdBBX7* RNAi transgenic plants treated with or without low temperature. RNA-seq analysis revealed that 6254 genes were up-regulated and 5831 genes were down-regulated in GL-3 following an 8-hour low-temperature (4°C) treatment (Supplementary Fig. S11A;

Supplementary Data Sets S1, S2). Under cold conditions, MdBBX7 appeared to regulate the expression of 8059 genes, with 3782 genes down-regulated and 4277 genes up-regulated in *MdBBX7* RNAi plants compared to GL-3 (Supplementary Fig. S11A; Supplementary Data Set S3). To investigate how MdBBX7 enhances cold tolerance in apple, we performed a Venn diagram analysis to identify cold-responsive genes regulated by MdBBX7. This analysis identified 1706 cold-induced genes that were down-regulated in *MdBBX7* RNAi plants under cold stress (Supplementary Fig. S11B; Supplementary Data Set S4) and 971 cold-repressed genes that were up-regulated in *MdBBX7* RNAi plants under cold stress (Supplementary Fig. S11C; Supplementary Data Set S5). These findings suggest that MdBBX7 may enhance cold tolerance by regulating the expression of these cold-responsive genes. Gene ontology (GO) enrichment analysis revealed that 1706 DEGs were enriched in biological processes related to response to cold, regulation of signaling, and fatty acid metabolic process (Supplementary Fig. S11D). 971 DEGs were enriched in biological processes related to monocarboxylic acid metabolic process, response to salicylic acid, and protein phosphorylation (Supplementary Fig. S11E). Among the 1706 DEGs, genes involved in cold response, including *MdCBF2* (Zhao et al. 2016), *MdCBF4* (Siddiqua and Nassuth 2011), and *MdCAMTA3* (Doherty et al. 2009), as well as genes related to fatty acid metabolism, including *MdFAD4* (Gao et al. 2009), *MdFAD5* (Heilmann et al. 2004) and *MdFAD8* (Tovuu et al. 2016) were identified. RT-qPCR analysis showed that MdBBX7 positively regulated the expression of these genes under cold stress (Fig. 7A, B; Supplementary Fig. S12A), consistent with the RNA-Seq data.

MdBBX7 is a typical B-BOX (BBX) transcription factor, recognizing the T/G-boxes, ACGT motif, and CCTTG motif in the promoters of target genes (Datta et al. 2007; Xu et al. 2018; Chen et al. 2022). Promoter sequence analysis showed that there are multiple binding elements of MdBBX7 in the promoter regions of *MdCBF2*, *MdCBF4*, *MdCAMTA3*, *MdFAD4*, and *MdFAD8*. Chromatin Immunoprecipitation (ChIP)-qPCR and electrophoretic mobility shift (EMSA) confirmed that MdBBX7 directly bound to *MdFAD8* and *MdCBF2* promoters under cold stress (Fig. 7C, D). However, no significant enrichment was observed in the promoters of *MdCBF4*, *MdCAMTA3*, and *MdFAD4* (Supplementary Fig. S12B-D). These findings demonstrate that MdBBX7 directly

1 regulates *MdFAD8* and *MdCBF2* expression. Previous studies have reported that
2 elevated levels of polyunsaturated fatty acids (PUFAs) during cold acclimation can
3 improve the freezing tolerance, especially linoleic acid (18:2) and linolenic acid (18:3)
4 (Kuiper 1970; Hamada et al. 1998). We revealed that the PUFA percentage was lower
5 in *MdBBX7* RNAi plants but higher in *MdBBX7* OE plants compared to GL-3 plants,
6 both before and after cold acclimation (Fig. 7F). In conclusion, these results indicate
7 that *MdBBX7* positively regulates cold tolerance through regulating cold-responsive
8 gene expression and FA desaturation.

9 **The cold-induced transcriptional activity of *MdBBX7* is dependent on the Ser198** 10 **phosphorylation of *MdMIEL1* by *MdKIN10***

11 We next tested whether *MdKIN10*-mediated phosphorylation of *MdMIEL1* affects its E3
12 ligase activity and substrate binding. Firstly, we performed an *in vitro* ubiquitination
13 assay to examine the effect of *MdKIN10*-mediated phosphorylation of *MdMIEL1* on its
14 E3 ligase activity. Our results showed that the level of ubiquitinated *MdBBX7* was
15 comparable when it was incubated with *MdMIEL1* or *MdMIEL1*^{S198D}. This indicates that
16 *MdKIN10*-mediated phosphorylation of *MdMIEL1* does not alter its E3 ligase activity
17 (Supplementary Fig. S13A). Secondly, we conducted a pull-down assay to assess the
18 effect of *MdKIN10*-mediated phosphorylation of *MdMIEL1* on its binding to substrates.
19 The results demonstrate that both *MdMIEL1* and *MdMIEL1*^{S198D} exhibited comparable
20 interactions with *MdBBX7*. Therefore, *MdKIN10*-mediated phosphorylation of *MdMIEL1*
21 does not influence its binding to substrates (Supplementary Fig. S13B).

22 We then explored whether the phosphorylation of *MdMIEL1* mediated by *MdKIN10*
23 influences the protein level of *MdBBX7* *in vivo*. Initially, we found that there was no
24 interaction between *MdKIN10* and *MdBBX7* (Supplementary Fig. S14), thus ruling out
25 the possibility that *MdKIN10* directly affects the protein level of *MdBBX7*. Relative to the
26 *MdMIEL1* OE calli, the overexpression of *MdMIEL1*^{S198D} resulted in increased *MdBBX7*
27 protein levels under both control and cold conditions. Conversely, the inhibition of
28 *MdMIEL1* phosphorylation by *MdKIN10* led to reduced *MdBBX7* protein levels under
29 both conditions (Fig. 8A). These results indicate that *MdKIN10*-mediated

1 phosphorylation of MdMIEL1 at Ser198 promotes MdBBX7 protein accumulation in
2 response to cold stress.

3 MdBBX7 directly binds to the promoters of *MdFAD8* and *MdCBF2* and positively
4 regulates their transcription (Fig. 7). We next investigated whether this regulation is
5 associated with MdKIN10-mediated phosphorylation of MdMIEL1 at Ser198. We
6 performed Dual-LUC assays in *N. benthamiana* leaves with a LUC reporter gene driven
7 by the *MdFAD8* or *MdCBF2* promoter. Cold treatment markedly increased the luciferase
8 activity derived from the *pMdFAD8:LUC* or *pMdCBF2:LUC* (Fig. 8B, C). In addition, the
9 luciferase activity driven by the *MdFAD8* promoter was further enhanced with the co-
10 transformation of MdBBX7 under control and cold conditions (Fig. 8B). The
11 transcriptional activation of the *MdFAD8* promoter by MdBBX7 was significantly
12 suppressed in the presence of MdMIEL1 or MdMIEL1^{S198A}, with MdMIEL1^{S198A} showing
13 stronger inhibitory effects under control and cold conditions. Interestingly, MdKIN10
14 could partially alleviate the repression of *MdFAD8* by MdMIEL1, instead of
15 MdMIEL1^{S198A} under normal and cold conditions (Fig. 8B). A similar pattern for *MdCBF2*
16 regulation by MdKIN10-MdMIEL1-MdBBX7 module under cold stress, but not under
17 control conditions (Fig. 8C). Taken together, these results suggest that the transcript
18 activity of *MdBBX7* is dependent on the Ser198 phosphorylation of MdMIEL1 by
19 MdKIN10 in response to cold.

20 As the upstream factors of MdBBX7 in cold response, MdMIEL1 and MdKIN10
21 might also regulate the downstream genes of MdBBX7. The results indicate that
22 MdMIEL1 negatively regulates the expression of *MdFAD8* and *MdCBF2*
23 (Supplementary Fig. S15A, B), thereby inhibiting the accumulation of PUFAs in apple
24 under cold conditions (Fig. 8D). Conversely, MdKIN10 was found to promote the
25 expression of *MdFAD8* and *MdCBF2*, as well as the accumulation of PUFAs (Fig. 8E;
26 Supplementary Fig. S15C, D). Furthermore, we explored the impacts of MdKIN10-
27 mediated phosphorylation on the regulation of gene expression and PUFA
28 accumulation. We discovered that *MdMIEL1*^{S198D} OE resulted in elevated expression of
29 *MdFAD8* and *MdCBF2*, as well as accelerated PUFA accumulation, whereas
30 *MdMIEL1*^{S198A} OE showed reduced levels relative to the *MdMIEL1* OE (Supplementary

Fig. S16A-C). These results suggest that MdKIN10 and MdMIEL1 regulate *MdFAD8* and *MdCBF2* expression and PUFA accumulation under cold stress.

Discussion

MdMIEL1, an E3 ubiquitin ligase, is known to ubiquitinate the transcription factors including MdMYB1, Md308L, MdBBX37, and MdBBX7 in apple responses to abiotic stress (An et al. 2017, 2020, 2021; Chen et al. 2022). Thus, it is critical for apple to precisely control the level of MdMIEL1 to ensure proper plant development and facilitate appropriate responses to a spectrum of environmental stresses. MdMIEL1 functions as a negative regulator of drought and cold tolerance, with its protein stability decreased in response to drought and cold stress (An et al. 2020; Chen et al. 2022). However, the molecular mechanisms underlying the degradation of MdMIEL1 under drought and cold have not been fully elucidated. In this study, we focus on cold stress and demonstrate that MdKIN10-mediated phosphorylation of MdMIEL1 at Ser198 is crucial for its autophagic degradation and apple cold tolerance.

A previous study showed that applying a 26S proteasome inhibitor successfully prevented MdMIEL1 degradation under cold conditions (An et al. 2020), however, the E3 ligases responsible for this degradation are largely unknown. In this study, we found that cold stress decreased MdMIEL1 protein abundance, and MdKIN10 mediated this degradation through the autophagy pathway (Fig. 2, 4). Thus, MdMIEL1 should be degraded *via* both autophagy and UPS pathways under cold stress. It would be interesting in the future to investigate the mechanism by which MdMIEL1 is degraded through the proteasome pathway, as well as whether there is a connection between the autophagy pathway and the 26S proteasome pathway in the degradation of MdMIEL1.

Multiple autophagy substrates harbor an AIM motif (Hu et al. 2022; Liu et al. 2023). We also revealed that the interaction between MdMIEL1 and MdATG8i was dependent on the AIM motif in MdMIEL1. Mutations in the AIM motif disrupted both the MdMIEL1-MdATG8i interaction and MdMIEL1 degradation under cold conditions (Fig. 5A, C, and D). These findings suggest that the MdMIEL1-MdATG8i interaction is vital for the

1 degradation of MdMIEL1 in response to cold stress. Furthermore, we observed that the
2 AIM motif in MdMIEL1 is not universally conserved across various species
3 (Supplementary Fig. S2C), indicating that the autophagic degradation of MIEL1 may not
4 be a conserved mechanism in plants, which might be one of the answers to why MIEL1
5 is degraded by both UPS and autophagy pathways.

6 PTMs are crucial in determining substrate specificity in autophagy, with
7 phosphorylation being particularly significant (Marshall et al. 2015; Richter et al. 2016;
8 Zhan et al. 2018). For instance, the selective autophagy receptor DSK2 is
9 phosphorylated by BIN2 near its AIM domains, enhancing its interaction with ATG8 and
10 promoting the degradation of the autophagy substrate BES1 (Nolan et al. 2017).
11 Additionally, phosphorylation of the autophagy receptor OPTN by TBK1 increases its
12 binding to ubiquitin chains, facilitating selective autophagy of damaged mitochondria.
13 Similarly, the ubiquitin-binding adaptor protein TNIP1 is phosphorylated by TBK1 on
14 serine residues proximal to its LIR motif, enhancing its interaction with LC3s in animal
15 models (Richter et al. 2016; Zhou et al. 2022). In this study, we found that the
16 phosphorylation of Ser198 in MdMIEL1 by MdKIN10 was essential for its autophagic
17 degradation. Mutational analysis revealed that phosphomimic variants of MdMIEL1
18 displayed increased interaction with MdATG8i, whereas preventing phosphorylation
19 reduced this interaction (Fig. 5F-H). Furthermore, mutation of the Ser198 residue to
20 alanine reduced the colocalization of MdMIEL1 with MdATG8i in the cytoplasm (Fig. 5J;
21 Fig. S6E). Thus, our work suggests that MdKIN10-induced phosphorylation of MdMIEL1
22 at Ser198 enhances its interaction with the autophagy protein MdATG8i, leading to
23 subsequent autophagic degradation of MdMIEL1 in response to cold stress. Previous
24 studies have indicated that phosphorylation of flanking residues at the N-terminus of the
25 core AIM motif can increase the acidity of the AIM, thereby strengthening its affinity for
26 the basic pocket of ATG8 (Di Rita et al. 2018; Johansen and Lamark 2020). Importantly,
27 the Ser198 residue of MdMIEL1 is located near the N-terminus of the core AIM motif.
28 Therefore, it is possible that phosphorylation of Ser198 may enhance the acidity of the
29 AIM, thereby increasing its affinity for the basic pocket on MdATG8i.

SnRK1, an evolutionarily conserved energy-sensor kinase, plays a critical role in plant growth, development, and disease resistance (Chan et al. 2017; Li et al. 2020a, 2022; Sun et al. 2021). Recent research has demonstrated that TaSnRK1 α positively regulates cold stress responses by phosphorylating TaPAP6L, thereby increasing endogenous JA levels in wheat (*Triticum aestivum*) (Zhang et al. 2023). In this study, we demonstrate that MdKIN10 phosphorylates MdMIEL1 and mediates the expression of cold-responsive genes, as well as the FA desaturation, to enhance cold tolerance. These findings suggest that MdKIN10 may have a conserved function in regulating cold stress responses in plants. Intriguingly, we observed that cold stress primarily enhances MdKIN10 kinase activity rather than its transcriptional expression (Fig. 2D; Fig.S4A), whereas *TaSnRK1 α* displayed a significant induction in transcript levels following cold stress (Zhang et al. 2023). Therefore, KIN10 employs distinct mechanisms in response to cold stress in apple and wheat.

BBX proteins are zinc-finger transcription factors that play diverse roles in plant development, photomorphogenesis, and the response to various biotic and abiotic stresses, including cold stress. For example, AtBBX7 and AtBBX8 act downstream of the AtCRY2-AtCOP1-AtHY5 module to improve plant cold tolerance via the CBF-independent pathway (Li et al. 2021). In apple, MdBBX37 positively regulates JA-mediated cold tolerance through the MdJAZ-MdBBX37-MdICE1-MdCBF pathway (An et al. 2021). We found that MdBBX7 functions downstream of MdMIEL1 to promote freezing tolerance (Fig. 6; Supplementary Fig. S10). RNA-seq data reveals that MdBBX7 modulates a set of cold-responsive gene expression, including *MdCBF2*, *MdCBF4*, and *MdCAMTA3* (Fig. 7B; Supplementary Fig. S12A). Therefore, the regulatory function of MdBBX7 in apple diverges from that of AtBBX7, in terms of dependence on the CBF pathway.

Plants grown at low temperatures show an increased level of FA unsaturation. This increase allows functional membrane fluidity to be maintained and is essential for plant survival under cold conditions (Los and Murata 2004). In Arabidopsis leaves, transcript levels of *FAD8* increase with decreasing temperature (Gibson et al. 1994). *osfad8* disrupts acclimation to cold temperatures (Tovuu et al. 2016). In our study, we found

that MdBBX7 directly promoted the expression of *MdFAD8*, promoting FA desaturation under both control and cold conditions (Fig. 7A, C, and F). In Arabidopsis, BBX7 and BBX8 have been shown to enhance plant freezing tolerance by upregulating genes involved in the biosynthesis and metabolism of anthocyanins, flavonoids, aromatic compounds, and amino acid derivatives, as well as those responding to ultraviolet signals. Therefore, our results offered insight into the mechanisms by which BBX transcription factors contribute to the regulation of cold tolerance by mediating FA desaturation in apple. Additionally, we discovered that MdKIN10 is essential for apple survival under freezing stress by regulating PUFA desaturation. Given that C18:3 serves as a precursor for JA biosynthesis, we hypothesize that MdKIN10 may function similarly to TaMdKIN10 in modulating JA production (Shen et al. 2011). Notably, although MdBBX7, MdMIEL1, and MdKIN10 regulated *MdFAD8* expression under control and cold conditions (Fig. 7A; Fig.S15A, C), they failed to significantly regulate the cold-induced expression of *MdFAD8*. Therefore, it is inferred that the primary regulation of *MdFAD8* expression in response to cold stress might be mediated by other proteins.

In summary, we propose a model that cold-activated MdKIN10 promotes the phosphorylation of MdMIEL1 at Ser198. The phosphorylated MdMIEL1 is then recognized by the autophagy-related protein MdATG8i, facilitating its selective degradation within the autophagosome. The degradation of MdMIEL1 relieves the inhibitory effect on MdBBX7, leading to an enhanced percentage of PUFAs and increased expression of cold-responsive genes under cold stress (Fig. 8F). Our study reveals an intricate PTM 'cascade' of MdKIN10-MdMIEL1-MdBBX7 molecular module under cold stress in apple.

Materials and Methods

Plant material and growth conditions

Leaves of apple 'Golden Delicious' were used for gene cloning. Four-week-old tissue-cultured GL-3 (the progenies of Royal Gala, with high regenerative capacity) was used for genetic transformation. Tissue-cultured GL-3 and transgenic plants were subcultured

every 4 weeks on Murashige & Skoog (MS) medium (4.43 g/L MS salts, 30 g/L sucrose, 0.2 mg/L 6-Benzylaminopurine [6-BA], 0.2 mg/L 3-Indoleacetic acid [IAA], and 7.5 g/L agar, pH 6.0). Four-week-old transgenic plants and GL-3 were rooted on MS medium (2.23 g/L MS salts, 20 g/L sucrose, 7.5 g/L agar, 0.5 mg/L IAA, 0.5 mg/L Indole-3-butyric acid [IBA], pH 6.0) (Xie et al. 2018). Seven weeks later, the plants were transplanted into the soil. Plants were grown under long-day conditions (16 h light/8 h dark) at 25°C.

Apple calli (*Malus × domestica* cv. Orin) were grown on MS medium (4.43 g/L MS salts, 30 g/L sucrose, 8 g/L agar, 0.4 mg/L 6-BA, 1.5 mg/L 2,4-D, pH 5.8) in darkness at 25°C. 10-day-old apple calli were used for gene transformation as described previously (Zhou et al. 2017).

Genetic transformation

To generate apple plants overexpressing *MdKIN10*, the coding region (CDS) of *MdKIN10* was cloned into a pK2GW7 vector. To obtain *MdKIN10* RNAi transgenic plants, a 171-kb fragment was inserted into the pK7WIWG2D vector. The resulting vector was transformed into *Agrobacterium tumefaciens* strain EHA105 and then transformed into GL-3 plants by an *Agrobacterium*-mediated transformation as described previously (Niu et al. 2022).

To obtain *MdKIN10* OE/*MdMIEL1* OE transgenic calli, the CDS of *MdKIN10* was cloned into a pK2GW7 vector. The resulting vector was transformed into *A. tumefaciens* strain EHA105, then transformed into *MdMIEL1* OE (with the hygromycin resistance) transgenic calli by an *Agrobacterium*-mediated transformation as described previously (Niu et al. 2022).

To obtain *MdMIEL1*^{S198A} OE and *MdMIEL1*^{S198D} OE transgenic calli, the Ser198 residue of *MdMIEL1* was mutated to alanine (A) or aspartic acid (D). These modified sequences were then cloned into a modified pCAMBIA1300 vector, which contains a hygromycin resistance gene. The resulting vector was transformed into *A. tumefaciens* strain EHA105, then transformed into wild-type 'Orin' apple calli by an *Agrobacterium*-mediated transformation.

Cold Stress treatment and the determination of electrolyte leakage

The 2-month-old plants were subjected to 4°C for 7 days for cold acclimation. To examine the survival rate under freezing conditions, non-cold acclimated and cold acclimated plants were treated with -10°C for the indicated time in the cold chamber (Thermo Fisher, BPHJS-120B). After treatment, the plants were recovered at normal conditions. Four weeks later, the survival rates were counted.

Cold tolerance of apple calli was evaluated as previously described with minor modifications (Shen et al. 2023). To examine the relative growth rates of calli under cold treatment, equal weights of WT and transgenic calli were grown on MS medium at room temperature for 10 days. Subsequently, they were treated at 4°C for an additional 10 days. Following this treatment, the relative growth rates were calculated.

Electrolyte leakage of leaves was detected as described previously (Guan et al. 2013; Xie et al. 2018).

Measurement of fatty acids

To determine the FAs of apple plants, two-month-old plants were treated with or without cold stress at 4°C for 9 days. The mature leaves were then collected. To determine the FAs of calli, calli were treated at 4°C or kept at 25°C for 7 days, and calli were collected after treatment. The mature leaves and calli collected above were thoroughly ground into powder and then put into a vacuum freeze dryer at -80°C for freeze drying (Labconco, USA, Freezone). FAs were extracted as previously described with slight modification (He et al. 2023). 0.1000 g frozen dried leaf powder (three duplications) was accurately weighed by 1/10,000 balance. FAs were extracted with 6 ml methanol solution containing 2.5% (v/v) H₂SO₄ at 80°C for 2 h. Then cooling to room temperature, the FA methyl esters were extracted with 1 mL hexane and 2 mL 0.9% (w/v) NaCl, and the organic phase was analyzed by gas chromatography, using methyl heptadecanoate as an internal standard. The GC-2014 instrument (Shimadzu) was equipped with a flame ionization detector and a 30-m (length) × 0.25-mm (internal diameter) × 0.5-μm (liquid membrane thickness) column (Supelco wax-10, Cat. no. 24079; Supelco). The initial column temperature was maintained at 160°C for 1 min, then increased by 4°C min⁻¹ to 240°C and held for 16 min at the final temperature. The peaks of each FA

composition were identified by their unique retention times, and their concentrations were calculated against the internal control.

Sequence alignment and phylogenetic analysis

The sequences for MdKIN10 homologs in other species were identified from the National Center for Biotechnology Information (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>) by performing a Basic Local Alignment Tool for Protein search with the MdKIN10 protein sequence as a query. Domain prediction was performed using the SMART program (<http://smart.embl-heidelberg.de/>). Protein sequences were aligned using the MUSCLE method in the software MEGA5.2 (https://www.megasoftware.net/older_versions). The phylogenetic analyses were conducted using MEGA version 5 and the Maximum Likelihood method with 1,000 bootstrap replicates.

Subcellular localization

Fluorescence was observed using a confocal laser scanning microscope (LEICA TCS SP8, Germany). For subcellular localization of MdKIN10, the CDS of *MdKIN10* were cloned into the pEearleyGate104 vector. The ER marker, WAK2-mCherry-HDEL, was created by combining the signal peptide of *Arabidopsis thaliana* wall-associated kinase 2 (AtWAK2) at the N-terminus of the fluorescent protein and the ER retention signal His-Asp-Glu-Leu (HDEL) at its C-terminus (Nelson et al. 2007). The resulting construct and ER marker were transformed into *Agrobacterium* strain GV3101, the *Agrobacterium* was finally diluted to optical density (OD) 600 = 1 and injected into *N. benthamiana* epidermis using a needle tube. The infiltrated *N. benthamiana* plants were grown for an additional 3 d to observe fluorescence. To image YFP fusions together with RFP fusions, samples were excited using a solid-state laser at the wavelength of 514 nm for YFP and a laser at 552 nm for mCherry. YFP fluorescence was detected with a 522- to 560-nm band-pass filter, and mCherry fluorescence was detected with a 590- to 640-nm band-pass filter. All imaging was performed by sequential scanning to avoid crosstalk between different colors. The methods used to analyze the colocalization of YFP-MdATG8i with mCherry-MdMIEL1 or mCherry-MdMIEL1^{S198A} were similar to those described above.

1 Y2H screening assay

2 A Y2H screening assay was carried out according to the manufacturer's manuals
3 (Clontech, 630439, 630489). For the Y2H screening, the zf-CHY domain (1-141 aa) of
4 *MdMIEL1* was cloned into pGBKT7 as bait. The cDNA library prepared from apple
5 leaves and roots served as prey. For the point-to-point Y2H assay, the CDS of *MdMIEL1*
6 and *MdBBX7* were cloned into pGADT7, and the CDS of *MdKIN10*, *MdATG8c*,
7 *MdATG8f*, and *MdATG8i* were cloned into pGBKT7, pGBKT7- and pGADT7- constructs
8 were transformed to yeast strains Y2H Gold. pGBKT7-53 and pGADT7-T were used as
9 positive controls. pGBKT7-Lam and pGADT7-T were used as negative controls. All
10 yeast strains were grown on synthetic double drop-out (SD/-Leu/-Trp) and synthetic
11 quadruple drop-out (SD/-Leu/-Trp/-Ade/-His) solid medium.

12 Pull-down assay

13 *In vitro* pull-down assay was carried out as previously described (Ju et al. 2022). In
14 brief, purified MBP-MdMIEL1/MdMIEL1^{S198D}/MdMIEL1^{mAIM} fusion protein were
15 incubated with MBP beads (cat. no. E8021S; NEB) at 4°C for 2 h, then incubated with
16 GST-MdATG8i or GST-MdBBX7-N214 (1-214 aa) for another 2 h. The mixture was
17 collected by centrifugation at 700 g for 1 min and subsequently washed five times with
18 PBS buffer. Proteins were separated on 10% (w/v) SDS-PAGE and detected with anti-
19 GST (cat. no. M20007; Abmart) and anti-MBP (cat. no. AE016; Abclonal) antibodies.

20 Immunoblot analysis of protein abundance

21 Protein was extracted from leaves or calli, using PEB buffer (50 mM Tris-HCl pH 7.5,
22 150 mM NaCl, 2 mM EDTA, 10% glycerol, 1% NP-40, 5 mM DTT, 1 mM PMSF). After
23 15 min of incubation on ice, the tubes were centrifuged at 13,000 g at 4°C for 15 min.
24 Protein samples were subjected to SDS-PAGE. Anti-AMPK α (Thr172) (cat. no. 2535T,
25 Cell Signaling Technology), anti-actin (cat. no. M20009M; Abmart), anti-GFP (cat. no.
26 D110008; Sangon), and anti-FLAG (cat. no. MA9019; Abmart) antibodies were used for
27 immunoblotting. To separate ATG8 and ATG8-PE, a Urea-Tricine SDS-PAGE system
28 was utilized as described previously (Schägger and von Jagow 1987), and Anti-ATG8
29 (cat. no. AS14 2769; Agrisera) antibodies were subsequently used for immunoblotting.

Nuclear-Cytoplasmic Fractionation

The nuclear-cytoplasmic fractionation was performed according to the method previously described (Wang et al. 2011). A total of 0.2 g of harvested callus was ground into a fine powder in liquid nitrogen and homogenized in lysis buffer (20 mM Tris-HCl, pH 7.5, 20 mM KCl, 2 mM EDTA, 2.5 mM MgCl₂, 25% glycerol, 250 mM sucrose, and 5 mM DTT), supplemented with protease inhibitors. The mixture was incubated on ice for 5 minutes, and the homogenate was filtered through a double-layer Miracloth. After centrifugation at 1,500 × g for 10 minutes, the supernatant, representing the cytoplasmic fraction, was further centrifuged at 10,000 × g for 10 minutes at 4°C to obtain the final cytoplasmic fraction. Nuclear proteins were resuspended in NRBT buffer (20 mM Tris-HCl, pH 7.4, 25% glycerol, 2.5 mM MgCl₂, and 0.2% Triton X-100) and washed three times by centrifugation at 1,500 × g for 10 minutes. The final nuclear pellet was resuspended in lysis buffer. MED37A and histone H3 were used as cytoplasmic and nuclear internal references, respectively, to validate the quality of the fractionation process.

Co-IP assay

The Co-IP analysis was performed according to the method previously described (Chen et al. 2022), with slight modifications. For the Co-IP assay of MdKIN10 and MdATG8i with MdMIEL1, the CDS of *MdMIEL1* was cloned into a modified pCAMBIA1300 vector, the CDS of *MdKIN10* or *MdATG8i* was cloned into vectors pGWB406. The resulting vectors were transformed into *Agrobacterium* strain GV3101 and co-transformed into *N. benthamiana* leaves. The infiltrated leaves were collected after being transformed for 3 days. Total proteins were extracted with extraction buffer and then incubated with an anti-GFP polyclonal antibody (cat. no. D110008; Sangon) overnight. Next, the protein-antibody complex was incubated with protein A/G agarose beads (cat. no. 26162; Thermo Fisher) for 3 h. After four washes with the elution buffer, the complexes were subjected to immunoblotting using anti-FLAG (cat. no. MA9019; Abmart) and anti-GFP (cat. no. D191040; Sangon) antibodies.

Split-luciferase complementation assay

The CDS of *MdKIN10* and *MdMIEL1* were cloned into vectors pCAMBIA1300-nLUC and pCAMBIA1300-cLUC, respectively. These constructs were then transformed into *Agrobacterium* strain GV3101 and co-transformed into *N. benthamiana* leaves. After growth for 72 h under a long-day condition, the LUC signal was captured using a Chemiluminescence Image Analysis System (Tanon 4600, China). The methods used to analyze the interaction of cLUC-MdATG8i with MdMIEL1-nLUC, MdMIEL1^{S198A}-nLUC or MdMIEL1^{S198D}-nLUC were similar to those described above.

BiFC assay

The CDSs of *MdKIN10* and *MdMIEL1* were cloned into vectors pSPYNE and pSPYCE. The resulting constructs were transformed into *Agrobacterium* strain GV3101, the *Agrobacterium* was finally diluted to optical density (OD) 600 = 1 and then co-transformed into *N. benthamiana* epidermis, confocal microscopy (LEICA TCS SP8, Germany) was used to examine the YFP fluorescence.

Protein kinase assays

The GRIK2-HIS, MBP-MdMIEL1/MBP-MdMIEL1^{S198A}-3XFLAG, and MBP-MdKIN10-KD recombinant proteins were purified from *E. coli* Rosetta-gami (DE3) pLys. *In vitro* phosphorylation assay was performed as previously described with minor modifications (You et al. 2023). To test the phosphorylation of MdKIN10 by GRIK2, ~2 µg HIS-GRIK2 and ~1.5 µg MBP-MdKIN10-KD were incubated in a total volume of 20 µl of kinase reaction buffer (20 mM Tris-HCl [pH 7.5], 10 mM MgCl₂, 0.1 mM CaCl₂, 2 mM dithiothreitol, and 50 µM ATP) containing 2 µCi [γ -³²P] ATP at 30°C for 30 min. The reaction was stopped by adding 5x SDS loading buffer. Proteins were boiled at 100°C for 5 min and then separated on 8% (w/v) SDS-PAGE. Proteins were stained by Coomassie brilliant blue. After 24 to 48 hours, the phosphorylated proteins were visualized by autoradiography (Typhoon 9410 imager).

To test the phosphorylation of MdMIEL1 by MdKIN10, ~1 µg GRIK2-HIS, ~1 µg MBP-MdKIN10-KD, and ~2 µg MBP-MdMIEL1-3XFLAG or MBP-MdMIEL1^{S198A}-3XFLAG were incubated in a total volume of 20 µL of kinase reaction buffer (20 mM Tris-HCl [pH 7.5], 10 mM MgCl₂, 0.1 mM CaCl₂, 2 mM dithiothreitol, and 50 µM ATP) at

30°C for 30 min. The reactions without ATP and the reactions with CIAP (cat. no. 2250A; Takara) were used as negative controls. The reaction was stopped by adding 5x SDS loading buffer. Proteins were boiled at 100°C for 5 min and then separated on 8% SDS-PAGE gels containing 50 µM Phos-tag and 100 µM MnCl₂. Phos-tag SDS-PAGE was performed following the manufacturer's protocol (cat. no. 304-93521; Wako). After the electrophoresis, the gel was washed three times with transfer buffer (50 mM Tris, 40 mM Glycine) for 10 minutes each time, with 10 mM EDTA added to the first two washes. The gel was then transferred to a PVDF membrane. Anti-FLAG antibody (cat. no. MA9019; Abmart) was used to detect the phosphorylation level of MdMIEL1.

For the *in vivo* protein kinase assay, total proteins were extracted from *MdMIEL1* OE, *MdMIEL1*^{S198A} OE, and *MdKIN10* OE/*MdMIEL1* OE calli and MdMIEL1 proteins were immunoprecipitated using an anti-FLAG antibody (cat. no. T61938S; Abmart), then MdMIEL1 phosphorylation signals were detected using an anti-phospho-Ser antibody (cat. no. P5747; Sigma). And Actin in total proteins was used as a control.

LC-MS/MS Analysis

Phosphorylated proteins were separated on 8% SDS-PAGE gels. After Coomassie brilliant blue R250 staining, the corresponding bands were cut into small pieces and decolorized. Then 50 mM NH₄HCO₃ solution containing 10 mM TCEP and 50 mM CAA was added to the dried protein granules, and the reaction was carried out at room temperature in the dark for 30 min. After removing the above alkylation reagent, 1 µg trypsin was added for overnight enzymatic hydrolysis, and the peptides were recovered by C18 column desalting and then entered into mass spectrometry for detection. The samples were analyzed by an Orbitrap Fusion Lumos mass spectrometer (Thermo Fisher Scientific) connected to an EASY-nLC 1200 system. The software Proteome Discoverer 2.5 (Thermo Fisher Scientific) was used for data analysis.

In vivo and *in vitro* ubiquitination

To perform the *in vivo* ubiquitination assay, four weeks old tissue-cultured GL-3 and *MdMIEL1* OE transgenic plants were treated at 4°C for 12 h. Total protein was extracted from GL-3, *MdMIEL1* OE transgenic plants and immunoprecipitated with an anti-

MdBBX7 antibody (GenScript, Nanjing, China), and was subjected to western blot analysis with an anti-ubiquitin (cat. no. PL026378; Abmart). Total protein was extracted from GL-3, *MdMIEL1* OE transgenic plants, Coomassie Brilliant Blue R-250 staining serves as a control.

For the *in vitro* ubiquitination assay, the fusion proteins of GST-MdBBX7-N214 and MBP-MdMIEL1 or MdMIEL1^{S198D} were incubated with ATP (cat. no. A1852; Sigma), Ubiquitin (cat. no. U-115; Boston BioChem), E1 (cat. no. E-304; Boston BioChem), and E2 (cat. no. E2-622; Boston BioChem) at 37°C for 4 h. The reaction products were then analyzed by immunoblotting using an anti-GST antibody.

RNA-seq, RNA Extraction, and RT-qPCR analysis

The 2-month-old *MdBBX7* RNAi transgenic plants and GL-3 were treated at 4°C for 0 or 8 h, mature leaves from two plants as one biological replicate were collected, and three biological replicates were used. Total RNA was extracted by a CTAB method as described previously (Cao et al. 2021). Construction of mRNA sequencing libraries and sequencing were performed on the Illumina HiSeq platform of Novogene (Beijing, China). Clean sequences were mapped to the *Malus × domestica* genome (GDDH13) by Hisat2. Differential gene expressions were analyzed by DESeq2 with a threshold of adjusted P values < 0.05 and the absolute value of log₂ (Fold change) > 0.5.

For RT-qPCR analysis, the first strand cDNA was synthesized with the RT SuperMix for qPCR (cat. no. R223-01; Vazyme). For qPCR, the reaction mixture was composed of a first-strand cDNA template, primer mix, and SYBR Green qPCR Premix (cat.no. EG201117M; iScience) in a final volume of 10 µL. The primers used are listed in Supplementary Data Set S6.

ChIP-qPCR assay

ChIP-seq assay was conducted as previously reported (Bowler et al. 2004; Xie et al. 2018), with slight modifications. Four weeks old tissue-cultured GL-3 were treated at 4°C for 12 h and then were used for crosslinking. Chromatin was purified and sonicated to shear DNA into 200-750 bp fragments. The chromatin solution was centrifuged and

the supernatant obtained was immunoprecipitated with an anti-MdBBX7 antibody or without antibody. Recovered DNA was used for ChIP-qPCR analysis. The primers are shown in Supplementary Data Set S6.

EMSA

EMSAs were performed as described (Chen et al. 2022). MdBBX7C which contained the last 204 aa (207-410 aa) in the C-terminal of MdBBX7, was cloned into pMAL-c5X (MBP). MBP-MdBBX7C was purified from *E. coli* Rosetta-gami (DE3) pLys. The probes were synthesized by Sangon Biotech Co., Ltd (Shanghai, China). EMSAs were performed using the LightShift Chemiluminescent EMSA Kit (cat. no. 89880; Thermo Scientific) as the manufacturer's protocol. The sequences of the *MdFAD8* and *MdCBF2* promoter probes are listed in Supplementary Data Set S6.

Dual-LUC reporter assay

The dual luciferase reporter assay was conducted as reported previously (He et al. 2024). The CDS of *MdBBX7*, *MdKIN10*, *MdMIEL1*, and point mutations of *MdMIEL1* were individually cloned into the pGreen62-SK vector. The promoter fragments of *MdCBF2* and *MFAD8* were introduced into the pGreen0800-LUC vector, respectively. The resulting constructs were individually transformed into *Agrobacterium* strain GV3101. The effector and reporter plasmids were mixed into different combinations and injected into 4-week-old *N. benthamiana* leaves. Firefly luciferase and Renilla luciferase activity were detected with a Dual-Luciferase Reporter Gene Assay Kit (cat. no. 11402ES60; Yeasen) by a multifunctional enzyme label instrument (Tecan, Switzerland). Six biological replicates for each experiment were used.

Statistical analysis

Significant differences were determined using Student's *t*-test; *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$. The statistical results are shown in Supplementary Data Set S7.

Accession Numbers

Sequence data from this article can be found in GDR under the following accession numbers: MdKIN10 (MD17G1053700), MdMIEL1 (MD16G1174300), MdBBX7 (MD17G1119500), MdGRIK2 (MD11G1134400), MdATG8i (MD16G1138400); and in NCBI under the following numbers: MdATG5 (XM_017322589.2), MdCBF2 (XM_008378874.3), MdCBF4 (XM_029106143.1), MdCAMTA3 (XM_029106983.1), MdFAD4 (XM_008387768.3), MdFAD5 (XM_008375743.3), MdFAD8 (NM_001293987.1), MdATG8c (XM_008366696.3), MdATG8f (XM_008349356.3).

Competing interests

The authors declare no competing interests.

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Author contributions

Q.G. and F.Z. conceived and designed the research; F.Z., T.F., Y.Z., and S.Z. performed the experiments; P.C., S.H., X.L., X.S., C.B., C.N. F.M., and Y.X. provided technical assistance; J.H. analyzed the data. Q.G. and F.Z. wrote the paper. All authors read and approved the manuscript.

Data availability

All data are available in the paper and its Supplementary Information. RNA-seq data have been deposited in the Genome Sequence Archive (GSA) at the Beijing Institute of Genomics (BIG) Data Center, Chinese Academy of Sciences, under the accession number PRJNA1195404.

Figure Legends

Figure 1. MdKIN10 interacts with MdMIEL1. **A)** MdKIN10 interacts with MdMIEL1 in a Y2H assay. The combination of pGBKT7-53 and pGADT7-T was used as a positive control, and the combination of pGBKT7-Lam and pGADT7-T was used as a negative control. AD and BD were the empty Y2H vectors. SD-L/W stands for media lacking Leucine and Tryptophan. SD-L/W/H/A stands for media lacking Leucine, Tryptophan, Histidine, and Adenine. **B)** Split luciferase complementation assay confirms the interaction between MdKIN10 and MdMIEL1 in tobacco leaves. **C)** Interaction between MdKIN10 and MdMIEL1 by Co-IP assay in tobacco leaves. Anti-GFP antibody was used for immunoprecipitation. Anti-GFP and anti-FLAG antibodies were used for immunoblotting. **D)** Interaction between MdKIN10 and MdMIEL1 by BiFC assay. MdMIEL1 was fused to the C terminus of YFP and MdKIN10 to the N terminus of YFP. WAK2-mCherry-HDEL was used as an ER marker. Scale bar, 25 μ m.

Figure 2. MdKIN10 phosphorylates and destabilizes MdMIEL1. **A)** MdKIN10 phosphorylates MdMIEL1 *in vitro*. The phosphorylation status of MdMIEL1 was examined using a Phos-tag gel assay (top panel). MdMIEL1-P, phosphorylated MdMIEL1. MdMIEL1, unphosphorylated MdMIEL1. CIAP: calf intestinal alkaline

phosphatase. The levels of MdMIEL1 (anti-FLAG), MdKIN10 (anti-MdKIN10), and MdGRIK2 (anti-HIS) proteins were shown on the SDS-PAGE gels. **B)** Analysis of phosphorylation sites of MdMIEL1 by activated MdKIN10 *in vitro*. The analysis was performed by LC-tandem mass spectrometry (MS/MS). **C)** MdKIN10 mediates the phosphorylation of MdMIEL1 at Ser198. The phosphorylation status of MdMIEL1 and MdMIEL1^{S198A} was examined in a Phos-tag gel assay (top panel). MdMIEL1-P, the phosphorylated MdMIEL1. MdMIEL1, the unphosphorylated MdMIEL1. The levels of MdMIEL1/MdMIEL1^{S198A} (anti-FLAG), MdKIN10 (anti-MdKIN10), and MdGRIK2 (anti-HIS) proteins were shown in the SDS-PAGE gels. **D)** MdKIN10 kinase activity in response to cold stress in apple. Two-month-old *M. domestica* plants were treated at 4°C for the time indicated, and the total proteins were used for the immunoblotting with anti-P-AMPKα (Thr172) antibody. Actin was used as a loading control. **E)** MdMIEL1 stability by MdKIN10 under cold stress. *MdMIEL1* OE, *MdKIN10* OE/*MdMIEL1* OE calli (10 days old) were treated at 4°C for 0, 6, or 12 hours. MdMIEL1 was detected with an anti-FLAG antibody, and actin was used as a control. **F)** Stability of MdMIEL1 and MdMIEL1^{S198A} under cold stress. The 10-day-old *MdMIEL1* OE and *MdMIEL1*^{S198A} OE calli were treated at 4°C for 0, 6, or 12 hours. MdMIEL1 was detected with an anti-FLAG antibody. Actin was used as a control. **G)** MdKIN10 mediates the phosphorylation of MdMIEL1 at Ser198 *in vivo*. The 10-day-old transgenic calli were treated at 4°C for 0, 6, or 12 h. MdMIEL1 was immunoprecipitated with an anti-FLAG antibody, and the phosphorylated MdMIEL1 were detected with a P-Ser antibody. Actin was used as a control. **H)** Phenotypes of the WT, *MdMIEL1* OE, *MdMIEL1*^{S198D} OE, and *MdMIEL1*^{S198A} OE transgenic apple calli after cold treatment. Apple calli (10 days old) grown on the medium were treated at 4°C for 10 days. WT, wild type. **I)** The relative growth rate of apple calli under cold treatment. Numbers below the blots indicate the relative protein levels calculated with ImageJ software. WT, wild type. Error bars indicate SD (n = 3 in (H)). Significant differences were determined using Student's *t*-test, *, *P* < 0.05; **, *P* < 0.01; ***, *P* < 0.001.

Figure 3. MdKIN10 is required for cold tolerance of apple. **A)** Whole-plant freezing tolerance of GL-3 and *MdKIN10* RNAi plants with or without cold acclimation. After the freezing treatment, all plants were transferred to the growth chamber for recovery. NCA,

1 Non-Cold Acclimated; CA, Cold Acclimated. Scale bars, 2 cm. **B)** Survival rates of GL-3
2 and *MdKIN10* RNAi plants in (A). **C)** Freezing tolerance of GL-3 and *MdKIN10* RNAi
3 plants determined by leaf electrolyte leakage assays. Six plants and two leaf discs from
4 each plant were used for each temperature point. **D)** Whole-plant freezing tolerance of
5 GL-3 and *MdKIN10* OE plants with or without cold acclimation. After the freezing
6 treatment, all plants were transferred to the growth chamber for recovery. Scale bars, 2
7 cm. **E)** Survival rates of GL-3 and *MdKIN10* OE plants in (D). **F)** Freezing tolerance of
8 GL-3 and *MdKIN10* OE plants determined by leaf electrolyte leakage assays. Six plants
9 and two leaf discs from each plant were used for each temperature point. Error bars
10 indicate SD (n = 3 in (B and E)), 27 plants were used and divided into three biological
11 replications; n = 6 in (C and F)). Significant differences were determined using Student's
12 *t*-test, **, $P < 0.01$; ***, $P < 0.001$.

13 Figure 4. *MdKIN10* promotes autophagic degradation of MdMIEL1. **A)** Degradation of
14 MdMIEL1 by *MdKIN10* through the autophagy pathway. FLAG-tagged MdMIEL1 was
15 transiently co-expressed with GFP or GFP-tagged *MdKIN10* in tobacco leaves. Protein
16 accumulation was examined by western blotting after 3 days of infiltration. MG132 or
17 E64d was applied 12 h before sample harvesting. Dimethyl sulfoxide (DMSO) was used
18 as a mock control, and actin served as a loading control. **B)** Autophagic degradation of
19 MdMIEL1 under cold. One-week-old calli expressing *GFP-MdMIEL1* were treated with
20 ConA and exposed to cold stress. GFP-MdMIEL1 fusion protein and free GFP levels
21 were detected using an anti-GFP antibody, with actin serving as the loading control. **C)**
22 Subcellular localization of YFP-MdMIEL1 in tobacco leaf protoplasts under control and
23 cold conditions. Tobacco leaves expressing *YFP-MdMIEL1* were treated with or without
24 1 μ M ConA for 12 hours before preparing protoplasts, followed by treatment under
25 control or cold conditions for 2 hours before imaging by confocal microscopy. Scale bar,
26 10 μ m. **D)** Quantification of YFP-MdMIEL1 puncta. **E)** The abundance of MdMIEL1 in
27 cytoplasm and nucleus in response to cold stress. *MdMIEL1* OE calli (10 days old) were
28 treated at 4°C for 0 or 12 hours. The protein contents in cytoplasm and nucleus were
29 separated. MdMIEL1 protein abundance was analyzed by immunoblotting using an anti-
30 FLAG antibody, and H3 and MED37A serving as marker proteins for the nuclear and
31 cytoplasmic fractions, respectively. **F)** Effect of autophagy pathway on the autophagic

degradation of MdMIEL1 in response to cold. *MdATG5* was transiently knocked down, and *GFP-MdMIEL1* was overexpressed in one-week-old calli. These calli were then treated with cold stress treatment. GFP-MdMIEL1 fusion protein and free GFP levels were detected using an anti-GFP antibody, with actin serving as the loading control. **G)** MdMIEL1 degradation by MdKIN10 *via* the autophagy pathway *in planta* under cold. GFP-MdMIEL1 was transiently expressed in WT and *MdKIN10* OE calli, then these calli were treated with ConA and exposed to cold stress. GFP-MdMIEL1 fusion protein and free GFP levels were detected using an anti-GFP antibody, with actin serving as the loading control. Numbers below the blots indicate the relative protein levels calculated with ImageJ software. WT, wild type. ConA, concanamycin A. Error bars indicate SD (n = 10 in (D)). Significant differences were determined using Student's *t*-test, ***, *P* < 0.001.

Figure 5. Ser198 phosphorylation of MdMIEL1 mediated by MdKIN10 is critical for its interaction with MdATG8i and autophagic degradation. **A)** Y2H assay of the interaction between MdMIEL1 or MdMIEL1^{mAIM} and MdATG8i. The combination of pGBKT7-53 and pGADT7-T was used as a positive control, and the combination of pGBKT7-Lam and pGADT7-T was used as a negative control. AD and BD were the empty Y2H vectors. SD-LW stands for media lacking Leucine and Tryptophan. SD-LW/H/A stands for media lacking Leucine, Tryptophan, Histidine, and Adenine. **B)** Split luciferase complementation assay between MdMIEL1 and MdATG8i in tobacco leaves. **C)** Pull-down assay between MdMIEL1^{mAIM} and MdATG8i. MdATG8i was fused to GST and MdMIEL1^{mAIM} to MBP. **D)** Analysis of GFP-MdMIEL1 and GFP-MdMIEL1^{mAIM} cleavage in response to cold treatment. GFP-MdMIEL1 or GFP-MdMIEL1^{mAIM} were transiently expressed in tobacco leaves, then tobaccos were subjected to cold treatment for 2 h. GFP-MdMIEL1/MdMIEL1^{mAIM} fusion protein and free GFP levels were detected using an anti-GFP antibody, with actin serving as the loading control. **E)** Schematic diagram of MdMIEL1 protein sequence showing the domains, AIM motif, and phosphorylation site. Blue text, MdKIN10 phosphorylation site; orange text, predicted MdATG8i interacting motif (AIM: W/L/Y-X-X-L/I/V). **F)** Interaction between MdATG8i and MdMIEL1 or MdMIEL1^{S198D} by the Pull-down assay. MdATG8i was fused to GST and MdMIEL1/MdMIEL1^{S198D} to MBP. **G)** Interaction between MdATG8i and MdMIEL1 or

MdMIEL1^{S198A} by the Co-IP assay. Anti-GFP and anti-FLAG antibodies were used for immunoblotting. **H)** Split luciferase complementation assay of MdMIEL1, MdMIEL1^{S198D}, or MdMIEL1^{S198A} with MdATG8i in tobacco leaves. **I and J)** The colocalization of YFP-MdATG8i with mCherry-MdMIEL1, mCherry-MdMIEL1^{mAIM} or mCherry-MdMIEL1^{S198A} in tobacco leaf protoplasts under cold. Protoplasts were treated under 2-h cold conditions before being imaged by confocal microscopy. Punctate structures in the cytoplasm (indicated by white arrows) are YFP-MdATG8i-labeled autophagosomes. Scale bar, 5 μ m.

Figure 6. MdBBX7 enhances cold tolerance of apple. **A)** MdMIEL1 mediates MdBBX7 degradation under cold stress. Four-week-old tissue-cultured GL-3 and *MdMIEL1* OE transgenic plants were treated with or without the proteasome inhibitor MG132 for 0 or 12 h under control or cold at 4°C. Total proteins were extracted and immunoblotted with anti-MdBBX7 antibody. Coomassie Brilliant Blue R-250 staining serves as a control. **B)** Whole-plant freezing tolerance of GL-3 and *MdBBX7* RNAi plants with or without cold acclimation. After the freezing treatment, all plants were transferred to the growth chamber for recovery. NCA, Non-Cold Acclimated; CA, Cold Acclimated. Scale bars, 2 cm. **C)** Survival rates of GL-3 and *MdBBX7* RNAi plants in (B). **D)** Freezing tolerance of GL-3 and *MdBBX7* RNAi plants determined by leaf electrolyte leakage assays. Six plants and two leaf discs from each plant were used for each temperature point. **E)** Whole-plant freezing tolerance of GL-3 and *MdBBX7* OE plants with or without cold acclimation. After the freezing treatment, all plants were transferred to the growth chamber for recovery. Scale bars, 2 cm. **F)** Survival rates of GL-3 and *MdBBX7* OE plants in (E). **G)** Freezing tolerance of GL-3 and *MdBBX7* OE plants determined by leaf electrolyte leakage assays. Six plants and two leaf discs from each plant were used for each temperature point. Numbers below the blots indicate the relative protein levels calculated with ImageJ software. Error bars indicate SD (n = 3 in (C and F, 27 plants were used and divided into three biological replications); n = 6 in (D and G)). Significant differences were determined using Student's *t*-test, *, *P* < 0.05; **, *P* < 0.01; ***, *P* < 0.001.

Figure 7. MdBBX7 facilitates FA desaturation and cold-responsive gene expression. **A)** The relative expression levels of *MdFAD8* in GL-3 and *MdBBX7* transgenic plants based on RNA-seq data. **B)** The relative expression levels of *MdCBF2* in GL-3 and

MdBBX7 transgenic plants based on RNA-seq data. **C and D)** ChIP assays showing the binding of *MdBBX7* to the *MdFAD8* and *MdCBF2* promoters *in vivo*. **E)** EMSA analysis showing that the *MdBBX7* protein binds to the promoters of *MdFAD8* and *MdCBF2*. **F)** Analysis of PUFAs contents in the leaves of GL-3, *MdBBX7* RNAi, and *MdBBX7* OE plants treated with control or cold at 4°C for 9 days. PUFAs represent the sum of C18:2 and C18:3. *PUFA percentage (%) = $(C18:2+C18:3)/(C16:0+C18:0+C18:1+C18:2+C18:3) \times 100\%$. Error bars indicate SD (n = 3 in (A, B, C, D, and F)). Significant differences were determined using Student's *t*-test, *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$.

Figure 8. The cold-induced transcriptional activity of *MdBBX7* is dependent on the Ser198 phosphorylation of *MdMIEL1* by *MdKIN10*. **A)** *MdBBX7* stability in *MdMIEL1* OE, *MdMIEL1*^{S198D} OE or *MdMIEL1*^{S198A} OE calli under control and cold conditions. Fourteen-day-old transgenic calli were treated at 4°C for 0 and 12 hours, then total proteins were extracted. *MdBBX7* protein was detected by anti-*MdBBX7* antibody, and actin was used as a control. **B and C)** The dual luciferase reporter assay showing the effects of *MdKIN10*, *MdMIEL1*, and *MdBBX7* on the expression of *MdFAD8* and *MdCBF2* before and after cold treatment. The protein and reporter construct were transiently expressed in tobacco leaves. The luciferase activities were normalized with Renilla luciferase as an internal control. **D and E)** PUFA contents in the leaves of GL-3 and *MdMIEL1* or *MdKIN10* transgenic plants under control or cold at 4°C for 9 days. **F)** Working models for the *MdKIN10*-*MdMIEL1*-*MdBBX7* under control and cold in apple. Under normal growth conditions, *MdKIN10* exhibits lower kinase activity, resulting in the accumulation of more non-phosphorylated *MdMIEL1*. This enhances the stability and abundance of *MdMIEL1*, which in turn promotes the ubiquitination and degradation of *MdBBX7*. Upon cold stress, *MdKIN10* is activated to phosphorylate *MdMIEL1* at Ser198, which promotes the interaction between *MdMIEL1* and *MdATG8i*, thereby targeting *MdMIEL1* for autophagic degradation. Consequently, *MdBBX7* is released from *MdMIEL1*, which then activates the transcription of *MdFAD8* and *MdCBF2*. This activation leads to an increased percentage of PUFAs and cold-responsive gene expression, thereby enhancing cold tolerance in apple. Numbers below the blots indicate the relative protein levels calculated with ImageJ software. Error bars indicate

SD (n = 8 in (B and C); n = 3 in (D and E)). Significant differences were determined using Student's *t*-test, *, *P* < 0.05; **, *P* < 0.01; ***, *P* < 0.001.

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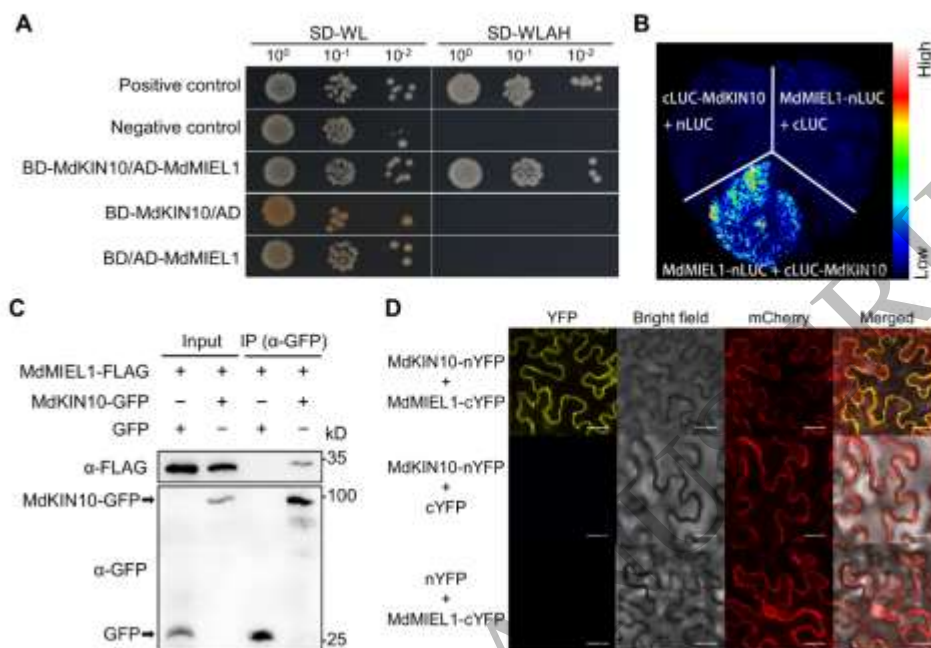


Figure 1

Figure 1
226x301 mm (x DPI)

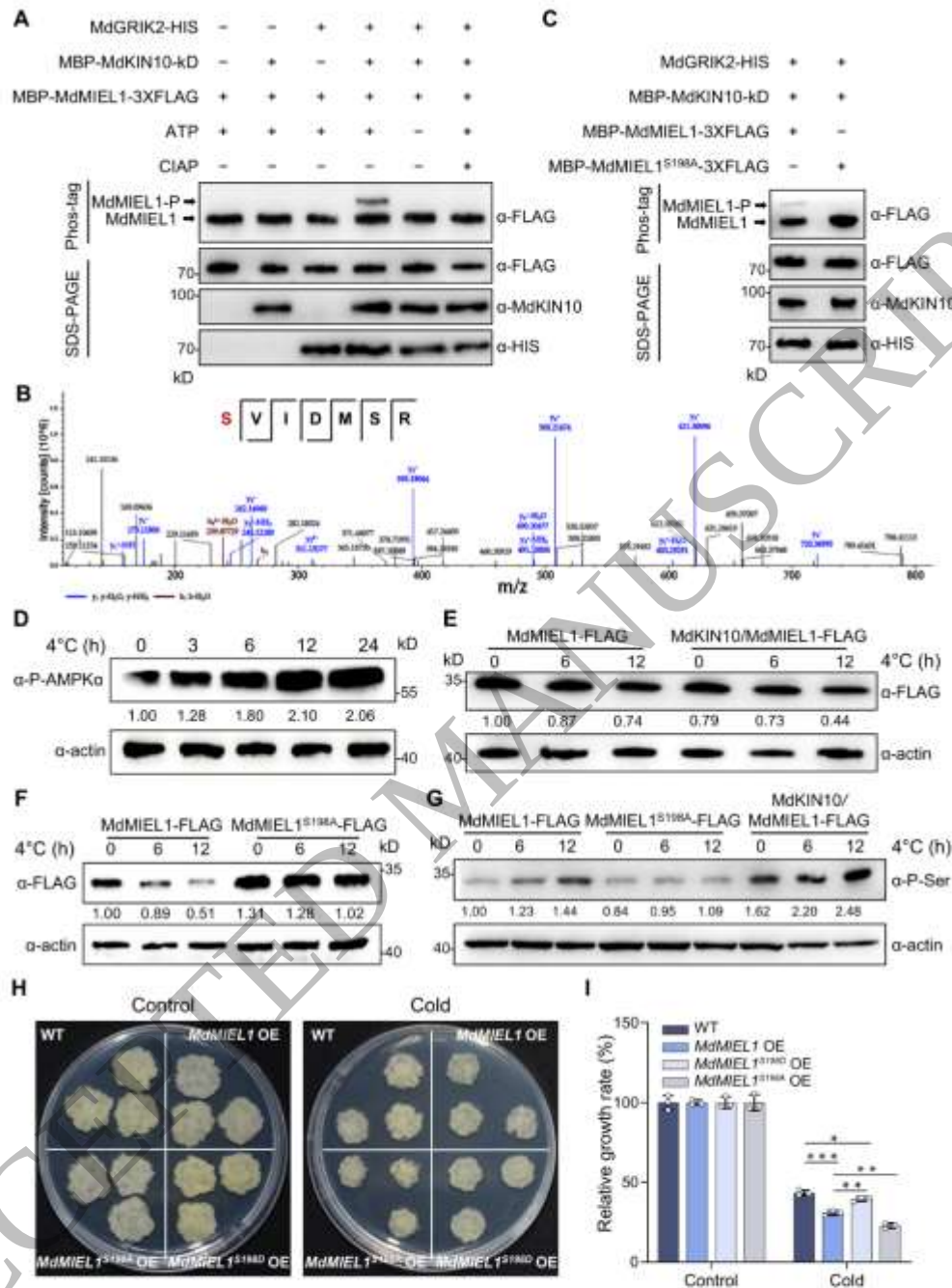


Figure 2

Figure 2
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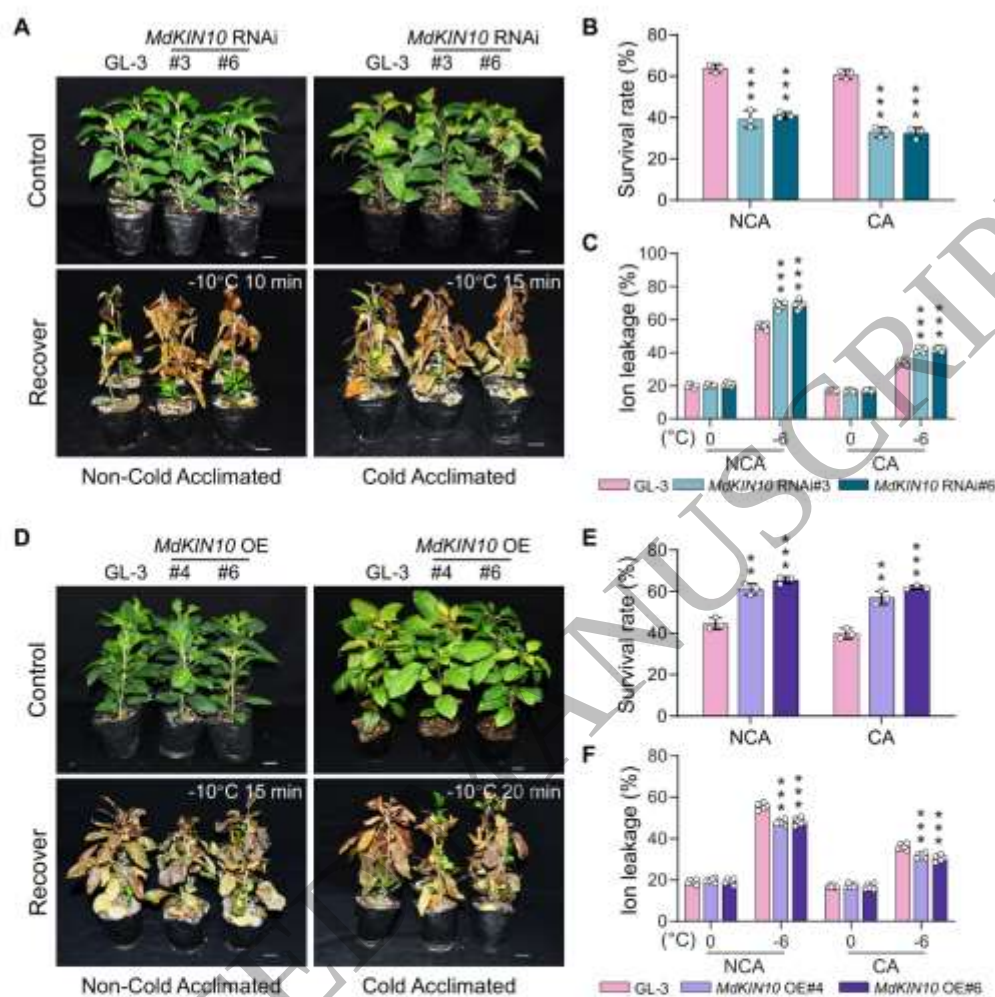


Figure 3

Figure 3
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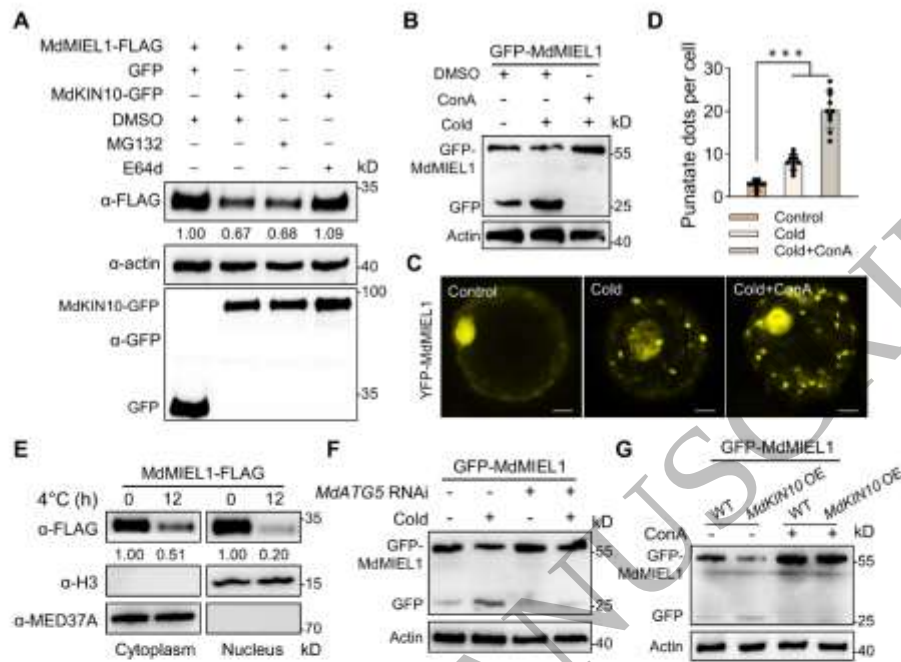


Figure 4

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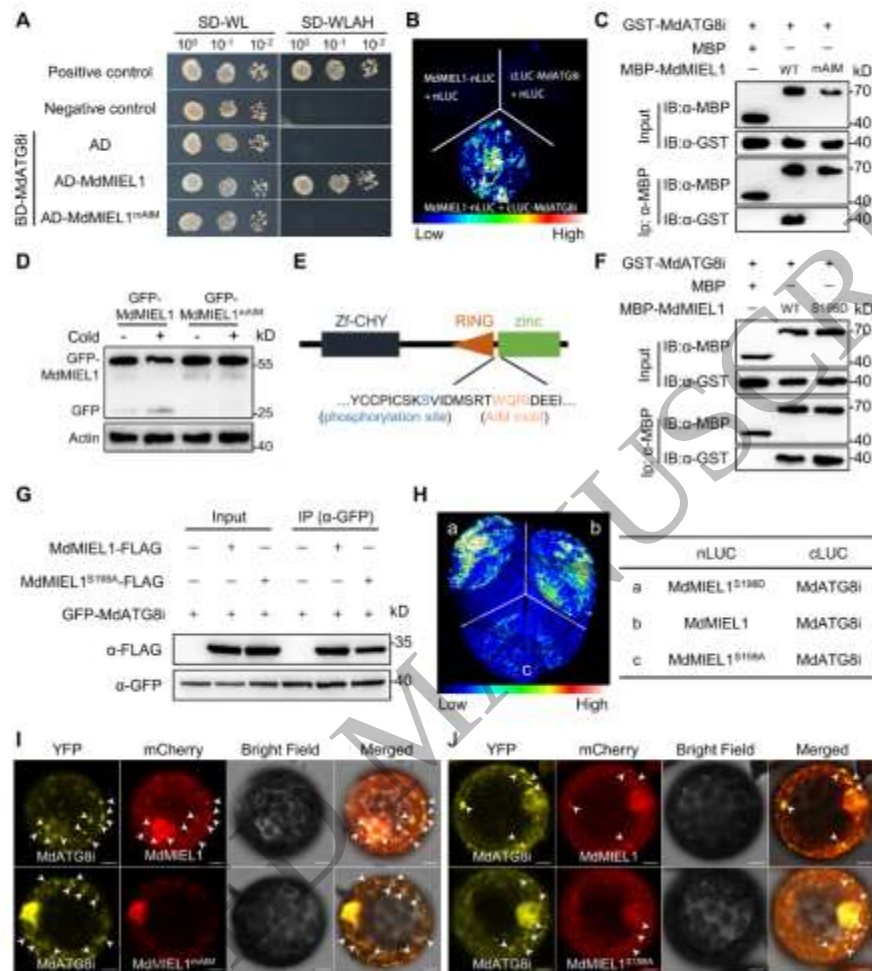


Figure 5

Figure 5
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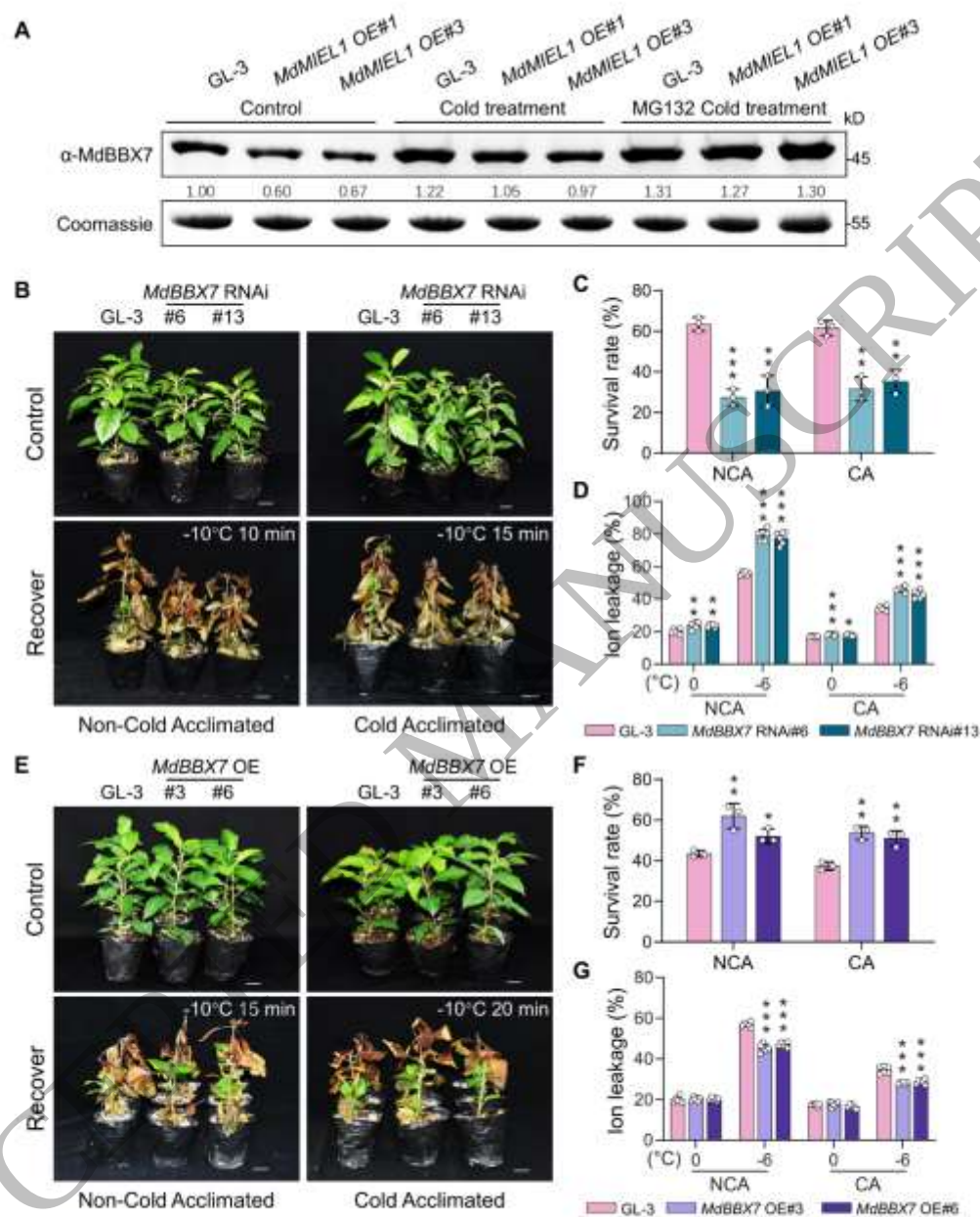


Figure 6

Figure 6
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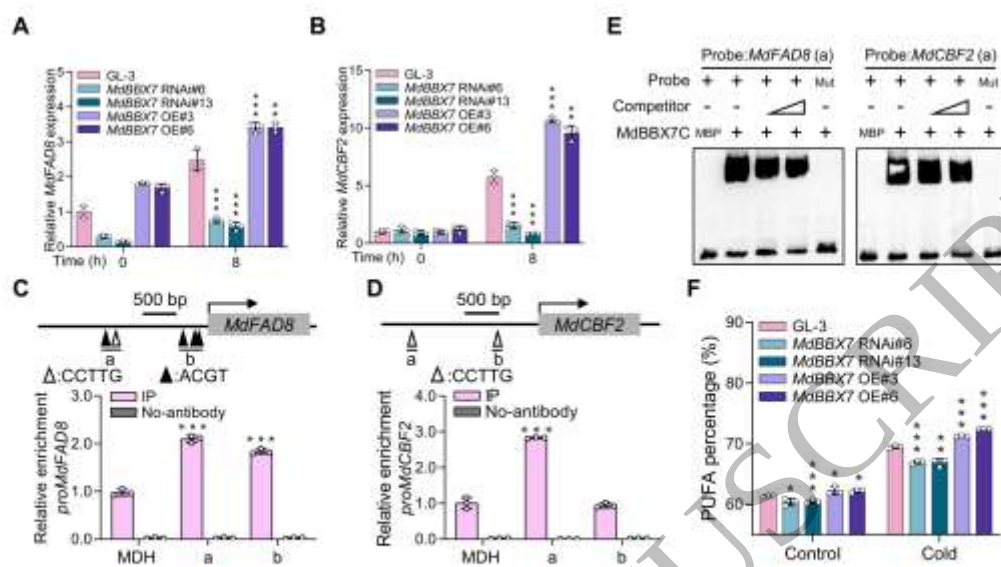


Figure 7

Figure 7
226x301 mm (x DPI)

